

MATHEMATICS GRADE 10: PROBABILITY I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 4.2 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 4.0: Statistics and Probability
Sub-Strand	Sub-Strand 4.2: Probability I
Total Duration	9 lessons × 40 minutes = 360 minutes total
Sub-Strand Content	– Experimental probability – Probability space – Mutually exclusive events – Independent events – Laws of probability (addition and multiplication laws) – Probability tree diagrams
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) carry out experiments to determine the experimental probability of events b) determine the probability space of an experiment and the probability of an event from it c) distinguish between mutually exclusive and independent events d) apply the laws of probability to solve problems involving mutually exclusive and independent events e) use probability tree diagrams to represent outcomes and solve problems involving combined events f) appreciate the application of probability in real life situations such as weather forecasting, farming and planning
Core Competencies	– Critical thinking and problem solving: learners reason about likelihood, compare experimental and theoretical results, and solve multi-step problems using the laws of probability and tree diagrams. – Communication and collaboration: learners work in groups to run experiments (coins, dice, bottle tops), share results and justify predictions about rainfall and other events. – Digital literacy: learners may use simulations or spreadsheets to run many trials and view how relative frequency settles. – Learning to learn: learners revise their probability model after each lesson, track the Driving Question Board and reflect on misconceptions.
Core Values	– Responsibility: learners record data honestly and use forecasts responsibly to protect crops, property and lives. – Integrity: learners report experimental results as observed, even when they differ from predictions. – Unity: learners cooperate in groups to carry out experiments and compare results. – Respect: learners listen to and consider the ideas of others when interpreting data.
Science & Engineering Practices	– Asking questions and defining problems: learners ask how likely a rain event is and what data would help answer it.

	– Planning and carrying out investigations: learners run repeated trials with coins, dice and spinners and record frequencies. – Analyzing and interpreting data: learners compute relative frequencies and compare them with theoretical probabilities. – Using mathematics and computational thinking: learners apply the addition and multiplication laws and tree diagrams to compute probabilities. – Developing and using models: learners build and revise a probability model of the weather forecast scenario across lessons. – Engaging in argument from evidence: learners justify decisions, such as when to plant, using probability values.
Pertinent & Contemporary Issues (PCIs)	– Disaster risk reduction: using probabilities of storms and floods to prepare communities in places such as Kisumu and the Tana River basin. – Food security: using rain forecasts to decide when to plant maize and beans or to irrigate. – Environmental education: understanding weather variability and climate change effects. – Safety and security: using storm probabilities to plan safe air and road travel.
Career Connections	– Meteorologist: uses probability to forecast rain, storms and drought at the Kenya Meteorological Department. – Farmer or agricultural extension officer: decides planting and irrigation times using rainfall probability, for example in Kericho or the Rift Valley. – Pilot or air traffic controller: uses storm probabilities for flight planning from JKIA. – Actuary or insurance analyst: uses probability to price risk and crop insurance. – Disaster management officer: plans for floods and storms based on likelihood of extreme weather.
Focus for Lessons	This unit uses weather forecasting as a context for learning how probability helps people deal with uncertainty and make informed decisions. Learners move from experiments and relative frequency to probability spaces, event types, the laws of probability and tree diagrams. Each lesson follows predict, observe, explain, DQB and model-building phases, steadily revising a model of how a forecast like '70% chance of rain' is made and used.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How is probability applied in real life situations? KICD KEY INQUIRY QUESTIONS: 1. How do we determine the probability of an event? 2. How do we apply probability in real life situations?
Anchoring Phenomenon	On a March morning before the long rains, the Kenya Meteorological Department forecasts a 70% chance of rain in Kericho and the Rift Valley. A maize farmer near Eldoret must decide whether to plant today or wait or irrigate. At the same time, a pilot at JKIA checks the storm probability before planning a flight to Kisumu, and the Nairobi county disaster office decides whether to prepare drainage and warn residents of possible floods. The atmosphere is complex and constantly changing, so meteorologists cannot say for sure what will happen; they express how likely a weather event is using probability. It is puzzling because the forecast is never certain: a 70% chance of rain still means it might stay dry, and a 30% chance might still bring rain. Why do farmers, pilots and cities trust and act on numbers that do not guarantee the outcome, and how are such numbers worked out and combined?
Storyline Thread	Lesson 1: Experimental probability I – Learners toss coins and bottle tops and predict how often 'rain' (heads) occurs; they record frequencies and meet relative frequency as a way to estimate likelihood, linking to forecasts built from past weather records. Lesson 2: Experimental probability II – Learners increase the number of trials, pool class data and see relative frequency stabilise; they use past rainfall records from Kericho to estimate the probability of rain. Lesson 3: Probability space I – Learners list all possible outcomes (sample space) for coins, dice and simple weather states (rain, no rain), and define events as subsets. Lesson 4: Probability space II – Learners calculate theoretical probability from the probability space, compare it with experimental

	results and explain the differences, revising the forecast model. Lesson 5: Mutually exclusive and independent events – Learners classify weather and everyday events (rain and sunshine at the same time; rain on Monday and Tuesday) as mutually exclusive, independent or neither. Lesson 6: Laws of probability I – Learners apply the addition law $P(A \text{ or } B)$ for mutually exclusive events to find the chance of rain or storm for a farmer's decision. Lesson 7: Laws of probability II – Learners apply the multiplication law $P(A \text{ and } B)$ for independent events to find the chance of rain on two consecutive days. Lesson 8: Probability tree diagrams – Learners draw tree diagrams for combined events like the weather over two days and compute the probabilities of the branches and outcomes. Lesson 9: Applying probability to real decisions – Learners use their complete model to advise the farmer, pilot and county disaster team, finalise the DQB and answer: How is probability applied in real life situations?
Supporting Phenomena	– A farmer in Kakamega must consider both the chance of rain on Monday and on Tuesday to decide when to plant, combining probabilities of several days. – A match official tosses a coin to decide who kicks off in a Harambee Stars game, a fair chance with two equally likely outcomes. – A shopkeeper in Mombasa notes the proportion of days when mandazi sell out to predict stock needs, using past data as experimental probability. – A county disaster team considers the chance of a flood and a road blockage occurring together, linking independent and mutually exclusive events.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (40 minutes (1 lesson)): Experimental Probability I: Coin Tosses and the 70% Forecast

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the phenomenon of the 70% rain forecast, let learners run their own coin or bottle-top trials, record frequencies and calculate relative frequency as a way of estimating how likely an event is, and open the Driving Question Board.
Knowledge	Learners will be able to: (1) describe probability as a measure of how likely an event is, from 0 (impossible) to 1 (certain), written as a fraction, decimal or percentage; (2) define a random experiment and a trial; (3) record frequencies from repeated trials in a tally table; (4) calculate experimental probability (relative frequency) = number of times the event occurred / total number of trials, e.g. $6/10 = 0.6$ for heads; (5) explain that the KMD forecast of 70% ($P(\text{rain}) = 0.7$) is a likelihood, not a guarantee.
Skills	NGSS SEP: Asking Questions and Defining Problems (generating questions for the DQB); Planning and Carrying Out Investigations (tossing, tallying, recording); Analyzing and Interpreting Data (tables and relative frequency); Developing and Using Models (initial model of how a forecast number is produced); Using Mathematics and Computational Thinking (fractions, decimals and percentages). Collaboration, communication and critical thinking.
Attitudes	Curiosity about why uncertain numbers guide real decisions; honesty in recording results exactly as they fall; respect for group members' ideas; appreciation of the work of the Kenya Meteorological Department and of farmers, pilots and county officers who use probability.
Key Inquiry Question	1. How do we determine the probability of an event? 2. How do we apply probability in real life situations? Driving question: How is probability applied in real life situations?
Purpose in Storyline	This is the opening lesson. It presents the anchoring phenomenon (a 70% chance of rain in Kericho and the Rift Valley, and the decisions of Wanjiru Chebet, Captain Otieno and Ms Achieng's county office). Learners use coin or bottle-top trials (Heads = rain) to meet relative frequency, the first tool for estimating likelihood. They see the Kericho record of 14 rainy days in 20 only as a teaser ($14/20 = 0.7$), and the DQB created here is revisited in every later lesson.
Safety Notes	Toss coins or bottle tops gently onto a desk or a cloth or exercise book placed on the desk so they do not roll away or fall on the floor. Ask learners to pick up any dropped objects immediately to avoid slipping. Keep coins out of mouths. Keep the noise low.

B. LESSON OVERVIEW

Mr Kamau begins by reading a KMD forecast: 70% chance of rain in Kericho and the Rift Valley on a March morning before the long rains. Learners meet three people who must act on it: Wanjiru Chebet, a maize farmer near Eldoret with 2 acres (plant today, wait or irrigate?), Captain Otieno, a pilot at JKIA planning a flight to Kisumu, and Ms Achieng of the Nairobi County disaster office (prepare drainage and warn residents?). Learners first predict what '70% chance of rain' means and whether it guarantees rain. In groups of 5 they then run a random experiment: toss a coin or bottle top 10 times (Heads = rain, Tails = no rain), tally the results and calculate relative frequency (for example 6 heads and 4 tails gives $6/10 = 0.6$). They notice that results vary from group to group and that no single toss can be predicted, yet the numbers still say something about likelihood. They are then told that KMD's number is worked out from many past days, and see a teaser: Kericho had rain on 14 of 20 past March days. The class builds the Driving Question Board with questions such as 'How is the 70% worked out?' and draws a first model of how a forecast number could be produced.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Random numbers for experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12  Search ARES for similar readings	Learners listen to the scenario: 'On a March morning before the long rains, KMD forecasts a 70% chance of rain in Kericho and the Rift Valley.' They meet Wanjiru Chebet (maize farmer near Eldoret, 2 acres), Captain Otieno (JKIA pilot flying to Kisumu) and Ms Achieng (Nairobi County disaster office). Individually (2 minutes), each learner writes in the notebook: (a) What do you think '70% chance of rain' means? (b) If the forecast says 70%, will it definitely rain? (c) If it says 30%, can it still rain? (d) Out of 10 tosses of a coin where Heads = rain, how many heads do you predict? Then they share with a partner and the class.	Read the forecast slowly and say: 'Imagine you are Wanjiru standing at the edge of your 2 acres of maize near Eldoret. The radio says seventy percent chance of rain. Do you plant today, wait, or irrigate?' Ask learners to write silently, then WAIT TIME of 10-15 seconds before any talk. Say: 'Turn to your partner and compare, but do not tell me the answer yet.' Cold-call 4 learners from different parts of the room: 'Tell us what your partner thinks 70% means.' Record predictions on the board WITHOUT saying whether they are correct. Cold-call 2 more learners for the coin prediction: 'How many heads out of 10, and why?' Probe: 'What makes you say that?' and 'Does everyone agree?' Record the range of predictions in a column titled 'Our predictions'.	Eliciting prior ideas through Predict-and-Compare: learners commit to ideas in writing before any instruction, so that later evidence can confirm, challenge or refine them. Recording competing meanings of '70%' on the board creates the puzzle (certain or not?) that drives the investigation and makes learners' own thinking the starting point.	Listen and note: Do learners treat 70% as a guarantee ('it will rain')? Do they link it to a count ('7 out of 10 days')? Do they use words like chance or likely correctly? Do they predict about 5 heads out of 10 and give a reason (two sides), or say it must be exactly 5? Tick a class list: guarantee view, frequency view, no idea. This identifies who needs support with the idea of likelihood during the investigation.
Observe Phase  VIDEO: Experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Theoretical and Experimental Probability Practice Source: CK-12  Search ARES for similar readings	Groups of 5 carry out the random experiment. Roles: tosser, tallier, recorder, checker, reporter (rotate). Each group tosses a coin or bottle top 10 times on a cloth or book on the desk. Heads (H) = rain (R); Tails (T) = no rain (N). They tally each trial in a table (Trial 1-10, Outcome, Tally, Frequency of H, Frequency of T) and calculate relative frequency = number of heads / total trials. They write it as a fraction, decimal and percentage, with one worked example from Mr Kamau: 6 heads and 4 tails gives $6/10 = 0.6 = 60\%$. They then compare their group's result with at least two other groups on the board table and notice differences. Finally, they see the teaser: 'KMD records show Kericho had rain on 14 of 20 past	Model one toss and tally on the board, then say: 'Before you toss, decide: who tosses, who tallies, who records, who checks the count, who reports?' Circulate and ask each group: 'What is a trial here?' and 'How many tosses so far?' Insist that results are recorded honestly: 'If it falls tails, we write tails, even if we hoped for rain.' After 10 tosses, show the worked example 6 heads, 4 tails: relative frequency of heads = $6/10 = 0.6$. Ask: 'What does the 0.6 tell us?' WAIT TIME of 10-15 seconds. Cold-call 3 groups to give their H count and relative frequency; write them in a class table. Ask: 'Why are the results not all the same? Could we know toss 11 before we toss it?' Cold-call 3 learners. Then reveal the teaser: 'KMD's records	Data-driven sensemaking with a Claim-Evidence structure: learners generate their own data, tabulate it and compare across groups. Seeing that each group's relative frequency differs (e.g. 0.6, 0.4, 0.5) while single tosses are unpredictable helps them build the idea that probability describes patterns over many trials rather than certainty about one trial. The Kericho teaser links the classroom experiment to the real forecast.	Observe: tally marks correct and counted to 10; frequencies of H and T add to 10; relative frequency written correctly as a fraction, decimal and percentage (e.g. $6/10 = 0.6 = 60\%$); learners record the outcome honestly. Check that learners convert $14/20$ to 0.7 without help. Note any learner who divides by the number of heads instead of the total trials, and re-teach: 'relative frequency = times it happened / total trials'. Ask each group's reporter to state in one sentence what their number means.

	March days; $14/20 = 0.7$.' Learners compute $14/20$ and notice it equals 0.7.	for Kericho show 14 rainy days out of 20 March days.' Ask learners to calculate 14 divided by 20 and compare it with 70%: 'What do you notice?' Do not yet explain how forecasts are built; say 'Keep that question for our board.' For bottle tops, say 'It may land unevenly; record what happens.'		
Explain Phase  VIDEO: Experimental versus theoretical probability simulation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12  Search ARES for similar readings	Learners use their data to build an explanation of the 70% forecast. In their groups they discuss and write 3-4 sentences answering: (1) What does a relative frequency of 0.6 tell us about the group's tosses? (2) Was it a certain result we could have known beforehand? (3) How could the number 0.7 from KMD be similar to the relative frequency we calculated? (4) Why might Wanjiru, Captain Otieno and Ms Achieng still act on a 70% forecast even though rain is not guaranteed? They share with the class and refine a class statement such as: 'A probability is a number between 0 and 1 that describes how likely an event is, estimated from what happened in many past trials.'	Pose: 'Our group got 6 heads out of 10. Could the next 10 tosses give exactly 6 again? Why or why not?' WAIT TIME of 10-15 seconds, then ask learners to talk in groups for 3 minutes. Circulate and prompt: 'What evidence from your table supports your claim?' and 'How is a forecast of 70% like your 0.6?' Cold-call 4 groups for their explanations. Press for precision: 'Does 70% mean it will rain, or that rain is likely?' Introduce vocabulary: random experiment, trial, frequency, relative frequency, probability scale from 0 (impossible) to 1 (certain). Draw the scale on the board with 0, 0.3, 0.5, 0.7, 1 and ask 3 learners to place 'it will rain in Kericho tomorrow (70%)' and 'the sun rises tomorrow'. Ask: 'Is a decision based on 70% foolish if it stays dry?' Cold-call 2 learners, and keep the open question: 'How is the 70% worked out?' for the board.	Collaborative explanation with Talk Moves (revoicing, adding on, agree/disagree): learners use their own data as evidence to explain what a probability number means. Connecting the classroom relative frequency to the forecast model builds the key idea that probability is a number from 0 to 1 estimated from repeated observations, while leaving gaps (how many trials? how are they combined?) that later lessons address.	Listen for: relative frequency described as 'times it happened out of all trials'; recognition that the single outcome is unpredictable; use of the 0-1 scale; connection to $14/20 = 0.7$. Ask learners to complete the exit statement: 'A 70% chance of rain means _____ but does not mean _____.'. Look for 'about 7 out of every 10 similar days have rain' and 'it will definitely rain'. Mark learners who say 70% means 70% of the area will be rained on, and revisit the meaning in the next lesson.
Driving Question Board (DQB) Creation  VIDEO: Theoretical and experimental probabilities Source: Khan Academy (English - US curriculum)	Learners help create the class Driving Question Board on chart paper with the central driving question: 'How is probability applied in real life situations?' Individually, each learner writes at least one question on a sticky note or paper strip, for example: 'How does KMD get 70%?', 'Why did the group get 0.6 and not 0.5?', 'What	Display the driving question and the two KICD inquiry questions: 'How do we determine the probability of an event? How do we apply probability in real life situations?' Say: 'What are you still wondering about the forecast, the coin tosses, or the farmer, pilot and county office?' Give 2 minutes of silent writing, then WAIT TIME	Question-driven inquiry using a class Driving Question Board: learners' own questions are sorted and displayed, giving ownership of the learning path. Categorising questions shows where the unit is heading (sample spaces, theoretical probability, laws, trees, decisions) and helps learners see what they know now and what	Check that each learner contributed at least one question; that questions relate to probability and the phenomenon; and that some go beyond recall (e.g., 'why' and 'how' questions). Note the number of questions on how the forecast is calculated, on combining events, and on decision-making. Use the sorted

 Search ARES for similar videos  EXERCISE: Theoretical and Experimental Probability Practice Source: CK-12  Search ARES for similar readings	if there is a storm as well as rain?', 'How does Captain Otieno use 10%?', 'What happens over two days?' They share in groups, then place their questions on the DQB under headings such as 'How is probability worked out?', 'How are probabilities combined?', and 'How do people use probability to decide?'.	of 10-15 seconds. Ask learners to share in groups and choose their best 2 questions. Cold-call 5 groups to read a question and say where it belongs on the board. Group similar questions together and ask: 'Which questions can we answer with more trials? Which need new ideas?' Keep the board visible in the classroom and tell learners they will return to it every lesson to cross off, answer and add questions.	remains to be figured out.	board to plan emphasis in later lessons.
Model Building Phase  VIDEO: Random numbers for experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Relative Frequency Interpretation Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners draw their first initial model (individually, then compare in groups) of how a forecast number like 70% could be produced and used. The model shows: past days observed (e.g., coin tosses or Kericho records) -> count how often the event happened -> relative frequency = times it happened / total trials -> probability estimate (0 to 1) -> decision by Wanjiru, Captain Otieno and Ms Achieng. They label examples: $6/10 = 0.6$ for their tosses and $14/20 = 0.7$ for Kericho. They list what is still uncertain on the model with '?' marks (e.g., how many trials are enough, how a storm fits in, how to combine two days). Finally, they complete the summary table.	Say: 'Draw a picture or flow chart that explains how a number like 70% could come from what happened on past days and how it helps Wanjiru, Captain Otieno and Ms Achieng.' Provide a starter frame: Evidence -> Calculation -> Probability -> Decision. Circulate and ask: 'Where does your 0.6 fit in your model?' and 'What part are you unsure about? Mark it with a question mark.' WAIT TIME of 10-15 seconds after each question. Cold-call 3 learners to share a model and 2 learners to name a '?' they still have. Tell the class: 'We will keep this model and change it in each lesson.' Collect models or keep them in notebooks. Close by revisiting the predictions from the start: 'Which predictions have changed, and why?'	Developing and Using Models: learners externalise their current understanding as a labelled flow diagram from evidence to decision. Marking uncertainties with '?' makes gaps explicit and sets up cumulative model revision over the unit as new ideas (more trials, sample spaces, theoretical probability, laws and tree diagrams) are added.	Look for models that include: repeated trials or past records; counting the event; relative frequency as a fraction or decimal; a link to the forecast number; and a decision. Success criteria: correct use of $6/10 = 0.6$ and $14/20 = 0.7$; clear '?' marks showing awareness of gaps. Common misconceptions to note: probability as a guarantee, using total heads instead of the fraction, or thinking each group's value must be equal to 0.5.

D. TEACHER REFLECTION

Which learners treated the 70% forecast as a guarantee at the start, and did the group results (varying around 0.5 to 0.6) shift their thinking? Did all groups record honestly and calculate relative frequency using total trials as the denominator? Which DQB questions show curiosity about how 0.7 is worked out, and which will be answered in Lessons 2-4? Were bottle-top results noticeably uneven, and did learners notice? What changes are needed to pacing, grouping or materials (coins and bottle tops) for the next lesson, which uses more trials and Kericho records?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We tossed a coin or bottle top 10 times in groups (Heads = rain, Tails = no rain) and recorded the results in a tally table, e.g. 6 heads and 4 tails. Different groups got different results. We also saw that Kericho had 14 rainy days out of 20 in past March records.
What did I learn?	Probability is a number from 0 (impossible) to 1 (certain) that tells us how likely an event is. Relative frequency (experimental probability) = number of times the event occurred / total number of trials, e.g. $6/10 = 0.6$ and $14/20 = 0.7$. A single trial cannot be predicted for certain.
How does this explain the phenomenon?	A forecast of 70% chance of rain does not guarantee rain. It is a likelihood estimated from past records, meaning that about 7 out of every 10 similar days have rain. Wanjiru, Captain Otieno and Ms Achieng can use such numbers to make decisions, but we still need to find out how the numbers are worked out and combined.

LESSON 2 (80 minutes (2 x 40-minute lessons)): Experimental Probability II: More Trials, Pooled Data and the Kericho Records

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners increase the number of trials by pooling class data, see that relative frequency settles near a steady value as trials grow, and use the Kericho rainfall record (14 rainy days out of 20) to estimate the probability of rain. This helps explain why a 70% forecast can be trusted even though it never guarantees rain.
Knowledge	By the end of the lesson, the learner should be able to: (1) calculate relative frequency = number of times the event occurred / total number of trials; (2) explain that as the number of trials increases, relative frequency becomes more stable (informal law of large numbers); (3) use pooled class data and the Kericho record to estimate $P(\text{rain})$ as an experimental probability.
Skills	Developing and using models (running relative frequency table and graph); analysing and interpreting data (comparing 10, 20, 40 and 80 trials); using mathematics and computational thinking (fractions, decimals and percentages); constructing explanations from evidence; communication and collaboration in groups of 5.
Attitudes	Curiosity about why numbers that do not guarantee an outcome are still trusted; patience and honesty in recording results exactly as they happen; respect for others' data; appreciation of how probability supports decisions by Kenyan farmers, pilots and county offices.
Key Inquiry Question	1. How do we determine the probability of an event? 2. How do we apply probability in real life situations?
Purpose in Storyline	Lesson 1 gave learners relative frequency from one small group of 10 tosses (6 heads, 4 tails, 0.6) and raised the 70% forecast puzzle. Lesson 2 shows that small samples wobble and larger pooled samples settle, so learners can see why a forecast built from many past days is more trustworthy than one built from a few. This sets up Lesson 3, where learners list all possible outcomes as a sample space.
Safety Notes	Toss coins and bottle tops on a desk or a cloth, not in the air above the head, so nothing hits a face. Pick up dropped coins at once so nobody slips. Keep bottle tops with sharp edges away from fingers. Groups tidy and return all materials at the end.

B. LESSON OVERVIEW

Mr Kamau reopens the puzzle: a KMD forecast says 70% chance of rain in Kericho and the Rift Valley, and farmer Wanjiru Chebet near Eldoret must still decide whether to plant, wait or irrigate. Learners recall their Lesson 1 group result (for example 6 heads out of 10, relative frequency 0.6) and predict whether a bigger number of tosses would give the same relative frequency. They then add data from every group to build a running relative frequency table: 10 tosses gives $6/10 = 0.6$, 20 gives $11/20 = 0.55$, 40 gives $22/40 = 0.55$, and 80 gives $42/80 = 0.525$. They plot these values and notice the changes become smaller as trials increase. Finally they analyse the Kericho March record (14 rainy days in 20 years, $14/20 = 0.7$) and see that a forecast of 0.7 is a relative frequency from past records. Learners leave with an evidence-based claim: more trials give a more stable and more trustworthy estimate of probability, and a single day can still surprise us.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Experimental	Learners recall their Lesson 1 group result (Heads = rain, 10 tosses, e.g. 6 heads and 4 tails,	Display the KMD forecast: '70% chance of rain in Kericho and the Rift Valley.' Say: 'Last lesson, one	Predict-Observe-Explain with a quick-write: committing to a written prediction before seeing data	Check notebooks for (a) a prediction with a reason, (b) correct statement of Lesson 1

probability Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12 Search ARES for similar readings	relative frequency 0.6). In groups of 5 they discuss: 'If our group tossed 80 times, would the relative frequency of heads still be 0.6? If every group in the class pooled results, what would happen?' Each learner writes an individual prediction in the notebook, sketches a quick guess graph of relative frequency against number of tosses, and notes whether they think the Kericho record of 20 March days will match the 70% KMD forecast.	group got 0.6 from 10 tosses. Wanjiru near Eldoret has to plant using a number like this. Would you trust a number from only 10 days?' Give individual WAIT TIME of 10-15 seconds before talking. Then cold-call 4 learners from different groups to share predictions and ask 'What makes you think so?' Record the range of predictions on the board without judging. Ask 2 learners to say what they think happens to the relative frequency as tosses increase, and note 'Predict: bigger sample = ?' on the chart.	creates cognitive tension and makes learners notice when their ideas are confirmed or challenged. The guess-graph makes their ideas about stability visible and testable.	relative frequency as heads/total, and (c) any language of 'more tosses' or 'closer to'. Note learners who think a small group's 0.6 must stay the same, or who think results will be exactly 0.5. Tally the prediction types (stays 0.6 / moves toward 0.5 / unpredictable) on the board to guide later discussion.
Observe Phase  VIDEO: Experimental versus theoretical probability simulation Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Theoretical and Experimental Probability Practice Source: CK-12 Search ARES for similar readings	Part A: Each group of 5 repeats 10 more tosses of a coin or bottle top (Heads = rain, Tails = no rain), combining with its Lesson 1 data where available, and records heads and tails in a tally table. Part B: The class pools data from 8 groups ($8 \times 10 = 80$ tosses) on a large chart and builds a running relative frequency table: 10 tosses, $6/10 = 0.6$; 20 tosses, $11/20 = 0.55$; 40 tosses, $22/40 = 0.55$; 80 tosses, $42/80 = 0.525$ (42 heads, 38 tails). Learners convert each fraction to a decimal and percentage and plot the four points on a graph of relative frequency (vertical) against number of tosses (horizontal), with a dotted line at 0.5. Part C: Learners study the Kericho record: last 20 years, same March date window, 1 day per year, rain on 14 days and no rain on 6 days, so relative frequency of rain = $14/20 = 0.7$. They compare this with the KMD forecast $P(\text{rain}) = 0.7$.	Model one tally and one calculation on the board: 'heads divided by total tosses.' Circulate and ask: 'How many trials so far? How many heads? What is your relative frequency?' Remind groups that results must be recorded honestly, even if they look odd. When building the pooled table say: 'Watch how the number changes as we add more tosses. Write down what you notice.' After the graph is plotted, WAIT TIME 10-15 seconds in silence, then ask: 'What do you notice about the jumps between points?' Cold-call 3 learners to describe the pattern. Then reveal the Kericho record: 'Wanjiru's forecast came from records like this: 14 rainy days out of 20. What is the relative frequency?' Cold-call 2 learners to compute and say what 0.7 means in words. Ask: 'Does 0.7 guarantee rain tomorrow?'	Data pooling and graph-based pattern noticing: combining small samples into a bigger sample lets learners experience the law of large numbers first-hand. The running table and graph make the shrinking changes visible, and the Kericho record connects the classroom experiment to a real forecast.	Observe whether learners (a) record tallies accurately, (b) calculate relative frequency correctly as a fraction, decimal and percentage, (c) plot points at 0.6, 0.55, 0.55 and 0.525 correctly, and (d) point out that the points get closer together or nearer 0.5 as tosses increase. Look for $14/20 = 0.7 = 70\%$ being computed correctly. Note any learner who divides by heads instead of by total trials.
Explain Phase  VIDEO: Random numbers	Groups use a Claim-Evidence-Reasoning frame to answer: 'Why is a relative frequency from 80	Give out the CER frame and say: 'Use numbers from our table. A claim without evidence is just an	Claim-Evidence-Reasoning with peer critique: structuring explanations around data forces	Look for explanations that (a) state that relative frequency becomes steadier or closer to a fixed value

for experimental probability Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Experimental probability Source: Khan Academy (English - US curriculum) Search ARES for similar readings	tosses (or 20 years of records) more trustworthy than one from 10 tosses?' Claim: what happens to relative frequency as trials increase. Evidence: the numbers 0.6, 0.55, 0.55, 0.525 and $14/20 = 0.7$. Reasoning: why small samples can wobble and bigger samples settle. They then discuss in pairs: 'If the Kericho record gives 0.7, what does a 70% chance of rain mean for a single day? Can it still stay dry?' Each group writes a short note for farmer Wanjiru explaining in plain language what 70% means and why the forecast can still be wrong on one day.	opinion.' Allow 3 minutes of group talk, then WAIT TIME 10-15 seconds before whole-class sharing. Cold-call 4 groups (different from the Observe Phase callers) to share claims and have the class respond with 'agree because' or 'disagree because'. Press for precision: 'What does stabilise mean here? Does it mean exactly 0.5?' Clarify that experimental probability estimates the true chance and is rarely exact (0.525 is close to 0.5). Do not give the theoretical explanation; say: 'We will explain why 0.5 later when we list all outcomes.' Ask: 'How does this help us explain the 70% forecast?' Collect two or three notes to Wanjiru and read them aloud.	learners to justify their ideas with numbers and to hear other perspectives. Writing for a real audience (Wanjiru) makes the abstract idea of probability meaningful and checks understanding of what a forecast means.	with more trials, (b) cite specific figures from the table, (c) say that 0.7 comes from past records (14 out of 20), and (d) accept that one day can still differ from the forecast. Misconceptions to note: 'the forecast was wrong because it did not rain', 'bigger samples always give exactly 0.5', or 'the next toss must balance out the last ones'.
Driving Question Board (DQB) Creation VIDEO: Experimental probability Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Use Simulations to Explore Experimental Probability (Flexbook) Source: CK-12 Search ARES for similar readings	Learners return to the Driving Question Board started in Lesson 1: 'How is probability applied in real life situations?' In groups they write on sticky notes (a) what they now know, for example 'more trials make relative frequency steadier' and 'Kericho record gives $14/20 = 0.7$', and (b) new questions, such as 'Why does the coin settle near 0.5?', 'How many trials are enough?', 'How do we list all possible outcomes?', or 'How do we combine rain on two days?'. They place notes under headings such as 'Answered so far' and 'Still to investigate' and link them to the farmer, pilot or county office.	Hold up the DQB and say: 'What evidence do we have today that helps with our driving question?' Allow WAIT TIME 10-15 seconds, then cold-call 3 learners to read one note and say where it goes. Group similar questions together, and highlight questions that link to upcoming lessons (e.g. 'Why exactly 0.5?' leads to sample spaces and theoretical probability). Ask: 'Captain Otieno checks a 10% storm chance. How might a number like that be worked out?' Add the question to the board. Tick off one question that the evidence has partly answered.	Driving Question Board as a living record: tracking questions and evidence over time shows learners that their understanding is growing and that their curiosity drives the unit.	Check that notes are specific and linked to evidence (e.g. mention 0.525 or $14/20$), that new questions go beyond recall, and that at least some questions point to the next ideas (all possible outcomes, combining days). Note groups that only restate facts and prompt them with 'What do you still wonder?'
Model Building Phase VIDEO:	Learners revise their developing 'Forecast Model' in notebooks or on a poster. They add: (1) experimental probability = number	Say: 'Add today's evidence to your model. What is new compared with Lesson 1?' Provide sentence starters: 'As the number of trials	Cumulative model revision: adding new evidence to a visible model helps learners connect each lesson to the anchoring	Check that models include the relative frequency formula, evidence that more trials give steadier values (with numbers),

Random numbers for experimental probability Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12 Search ARES for similar readings	of times the event occurred / total trials; (2) the running relative frequency table and a sketch graph showing values settling near a steady value; (3) the Kericho estimate $P(\text{rain}) = 14/20 = 0.7$, matching the KMD forecast; (4) a note that a larger number of trials gives a better estimate but never a guarantee for one day. They draw a simple diagram linking past records $\rightarrow$ relative frequency $\rightarrow$ forecast 70% $\rightarrow$ decisions by Wanjiru, Captain Otieno and Ms Achieng's county office. They write one sentence explaining why Wanjiru can plan with 0.7 even though it might stay dry.	increases, the relative frequency ...' and 'The forecast of 0.7 comes from ...'. Circulate and ask 3 learners: 'Where does your model show the effect of more trials?' Use WAIT TIME 10-15 seconds before they answer. Cold-call 3 learners to share their diagram and their sentence about Wanjiru. Close by asking: 'What part of our model is still missing?' and note 'a way to list all possible outcomes' for Lesson 3.	phenomenon and see probability as a toolkit that is built step by step.	the Kericho estimate $14/20 = 0.7$, and the idea that a probability helps decisions but does not guarantee one outcome. Use the exit ticket: 'Why can Wanjiru trust a 70% forecast even if it stays dry tomorrow?' Look for references to many past days and long-run frequency.
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D. TEACHER REFLECTION

Which learners recognised that 0.55, 0.55 and 0.525 are steadier than 0.6 from 10 tosses, and who still expected exactly 0.5? Did groups record honestly or adjust results to match expectations? Did the Kericho record ($14/20 = 0.7$) help learners connect the experiment to the forecast? Which misconceptions (for example the forecast being 'wrong' after a dry day, or the idea that tails is due after several heads) need revisiting in Lesson 3 and Lesson 4? What pacing changes would help ensure every learner calculates and plots at least one relative frequency?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We pooled results from 8 groups (80 tosses, 42 heads and 38 tails) and watched the running relative frequency of heads change: 10 tosses, 0.6; 20 tosses, 0.55; 40 tosses, 0.55; 80 tosses, 0.525. The Kericho March record showed rain on 14 days out of 20, so $14/20 = 0.7$.
What did I learn?	Relative frequency = number of times the event occurred / total number of trials. As the number of trials increases, relative frequency becomes steadier and gives a better estimate of probability. Pooled data and long records (like Kericho) give more reliable estimates than small samples.
How does this explain the phenomenon?	A forecast of 70% chance of rain is based on a long record of similar days (about 7 in every 10 had rain), so it is a trustworthy estimate. It does not guarantee rain on one day, but it helps Wanjiru, Captain Otieno and the county office make sensible decisions over many days.

LESSON 3 (40 minutes (1 lesson)): Probability Space I: Listing Sample Spaces and Defining Events

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners list all possible outcomes of a random experiment as a sample space $S = \{\dots\}$ and describe an event as a subset of the sample space, so that the forecast of rain in Kericho and the Rift Valley can be seen as a statement about one event within a complete list of possible day states.
Knowledge	By the end of the lesson, learners should be able to: (1) define a random experiment, an outcome, a sample space and an event as a subset of S ; (2) list the sample space for one coin $\{H, T\}$, two coins $\{HH, HT, TH, TT\}$, one die $\{1, 2, 3, 4, 5, 6\}$ and a weather day $S = \{\text{rain, storm, dry}\}$; (3) write events such as 'even number' = $\{2, 4, 6\}$ and 'rain for the farmer' = $\{\text{rain, storm}\}$.
Skills	Developing and using models (listing sample spaces systematically); Analysing and interpreting data (linking Lesson 1-2 toss records to outcomes in S); Using mathematics and computational thinking (set notation); Constructing explanations (why a complete list matters before assigning chances); Communication and collaboration in groups of 5.
Attitudes	Curiosity about how uncertain weather can be organised into a complete list of possibilities; Patience and accuracy in listing every outcome without repeating or missing any; Respect for group members' ideas; Appreciation of how mathematics supports decisions by farmers like Wanjiru Chebet, pilots like Captain Otieno and county offices.
Key Inquiry Question	1. How do we determine the probability of an event? (This lesson builds the first step: knowing all possible outcomes and which of them make up the event.) 2. How do we apply probability in real life situations?
Purpose in Storyline	Lessons 1-2 taught learners to do trials, record frequencies and calculate relative frequency as an estimate of likelihood, with Heads = rain. Lesson 3 asks: what are ALL the things that could happen? Learners build sample spaces for coins, a die and a weather day $S = \{\text{rain, storm, dry}\}$ and define events as subsets. This gives the complete list that Lesson 4 will use to calculate theoretical probability (favourable / total outcomes). Learners do not calculate theoretical probabilities here; they only list outcomes and describe events.
Safety Notes	Learners toss coins or bottle tops gently on desks or a mat so they do not roll away or trip anyone. Dice or number cards should be kept on the desk. Keep walkways clear during group work. Collect all coins and bottle tops at the end.

B. LESSON OVERVIEW

Anchoring phenomenon: On a March morning KMD forecasts a 70% chance of rain in Kericho and the Rift Valley. Wanjiru Chebet near Eldoret must decide whether to plant, wait or irrigate; Captain Otieno at JKIA checks the storm chance before flying to Kisumu; Ms Achieng's Nairobi county office decides on drainage and warnings. In Lessons 1-2 learners tossed coins (Heads = rain) and calculated relative frequencies, for example $6/10 = 0.6$ for one group and $42/80 = 0.525$ for the class, and met the Kericho record of 14 rainy days out of 20. Today Mr Kamau asks: 'Before KMD can say 70%, what must it know first?' Learners discover that forecasters must first list every possible thing that could happen on the day. They list outcomes for one coin, two coins and a die, then build the weather sample space $S = \{\text{rain, storm, dry}\}$. They then define events as subsets, such as 'even number' = $\{2, 4, 6\}$ and 'rain, as the farmer sees it' = $\{\text{rain, storm}\}$. They add to the Driving Question Board (DQB) and revise the class model: a forecast is about an event within a complete list of outcomes.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Sample spaces for compound events Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Sample spaces for compound events Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the Lesson 1-2 coin tosses (Heads = rain, Tails = no rain). Individually (2 minutes) then in groups of 5 (3 minutes), they respond to the prompt: 'Mr Kamau tosses TWO coins together. Predict and write down every different result that could appear. How many different results will there be?' Many predict 3 results (two heads, one head, no heads) or 4. Learners also predict: 'On the day of the forecast, what are all the different kinds of weather KMD must think about for Kericho?' They record predictions in their exercise books with a reason.	Display the forecast: 'KMD says 70% chance of rain in Kericho.' Say: 'Before KMD can give a number, what must it know first? Think quietly.' WAIT TIME 10-15 seconds before anyone speaks. Then say: 'Write your prediction for two coins. Give a reason, not just a number.' Circulate and ask 'What makes you say three?' Cold-call 4 groups (not volunteers) to share, using 'Say more about how you decided.' Record all predictions on the board without judging. Say: 'We will test these predictions today.'	Prediction with Justification (Predict-Observe-Explain). Committing to a written prediction with a reason exposes the common idea that HT and TH are the same result. Learners are invested in checking it, and the later contrast between prediction and evidence drives conceptual change.	Look for: (1) each learner has a written prediction and a reason; (2) at least 3 groups predict 3 results for two coins, and this is recorded as a misconception to test, not corrected; (3) learners mention rain and dry (and possibly storm or other weather) when listing weather possibilities. Tick on a class checklist those who give a reason.
Observe Phase  VIDEO: Subsets of sample spaces Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Sample spaces for compound events Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Activity 1 (12 minutes): Each group has a coin or bottle top (the coin is fair), and a die or a set of number cards 1-6. Station A: one coin. Learners toss 10 times and record each result, then list every different outcome seen: {H, T}. Station B: two coins (one coin in each hand, or a 5-shilling coin and a 1-shilling coin to tell them apart). Learners toss 20 times, recording results as ordered pairs (first coin, second coin) and tally which appear: HH, HT, TH, TT. They notice HT and TH are different when the coins are different. Station C: one die. Learners roll 12 times and list all faces: {1, 2, 3, 4, 5, 6}. They then mark which outcomes belong to the event 'even number'. Station D (paper-based): Learners read a KMD-style day card with states 'rain only', 'storm' and 'dry' (matching Wanjiru's, Captain Otieno's and Ms Achieng's concerns) and write $S = \{\text{rain,}$	Introduce the words: 'A random experiment is an action where we cannot be sure of the result. Each possible result is an outcome.' Model with one coin on the board. Say: 'Tails or heads: what other outcome is possible? If none, our list is complete.' For station B, ask: 'Are HT and TH the same? How could you tell the two coins apart?' Hold up two different coins. Prompt groups: 'How do you know you have not missed an outcome?' and 'Show me a systematic way: fix the first coin, then vary the second.' WAIT TIME 10 seconds after each question. For Station D say: 'Captain Otieno worries about storms. Is a storm the same as ordinary rain here? Why does KMD list it separately?' Cold-call 3 groups to read their lists aloud. Circulate to check that listing uses curly brackets and commas. Do not state any probabilities for the outcomes at this stage.	Structured Data Collection and Systematic Listing. Learners first generate outcomes by experiment (building on Lessons 1-2 trials) and then organise them with an ordered method (fix the first, vary the second). This moves them from 'what happened' in trials to 'what could happen' in a complete list, which is the basis of a sample space.	Observable indicators: (1) groups write $S = \{H, T\}$ and $S = \{HH, HT, TH, TT\}$ with curly brackets; (2) groups list HT and TH as two different outcomes and can explain why; (3) groups write $S = \{1, 2, 3, 4, 5, 6\}$ without repeating or missing a face; (4) groups write $S = \{\text{rain, storm, dry}\}$ for the weather day; (5) groups use a systematic method rather than random guessing. Note groups that list only outcomes they happened to see in their trials and prompt them: 'Is there an outcome that could happen but did not appear?'

	storm, dry}. They then circle the outcomes where 'rain' as the farmer sees it happens: rain or storm. Groups record in a table: Experiment Outcomes seen Complete list (S) Number of outcomes.			
Explain Phase  VIDEO: Sample spaces for compound events Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Probabilities of compound events <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Activity 2 (10 minutes): Learners compare their Observe Phase lists with their Predict Phase predictions and revise. In groups they then answer: (a) What is the sample space S for each experiment? (b) How many outcomes are in each? (c) Define an event as a subset: write the event A = 'even number' on a die as $A = \{2, 4, 6\}$; write the event 'at least one head' with two coins as {HH, HT, TH}; write the event 'rain, as Wanjiru sees it' as {rain, storm}; write the event 'storm for Captain Otieno' as {storm}. (d) Explain in 2-3 sentences: 'Why must KMD list every possible day state before saying 70%?' Learners also link to Lessons 1-2: in the coin tosses each toss was a trial that gave one outcome from S; the Kericho record of 14 rainy days and 6 days without rain were outcomes of the days observed. They complete a Claim-Evidence-Reasoning (CER) statement: Claim: 'An event is a part of the sample space.' Evidence: from their lists. Reasoning: why $\{2, 4, 6\}$ is a subset of $\{1, 2, 3, 4, 5, 6\}$.	Say: 'An event is a subset of S, a collection of outcomes that we are interested in.' Draw a Venn-style box on the board for $S = \{1, \dots, 6\}$ with $\{2, 4, 6\}$ circled. Ask: 'Is every outcome in the event also in S? Can an event have just one outcome? Could an event be empty or be all of S?' WAIT TIME 10-15 seconds. Cold-call 5 learners (one per group) to state an event in words and write its subset. Use the sentence frame: 'The event ____ is the subset {____} of S.' Press for reasoning: 'Wanjiru says rain. Captain Otieno says storm. Same event or different? Why does it matter that they define the event differently?' Highlight that the outcomes (rain only, storm, dry) are single results, while 'rain for the farmer' groups two outcomes. Correct the earlier misconception about two coins by returning to the board predictions: 'Which predictions were missing an outcome? What did the coins show us?' Avoid introducing probability values for events; say: 'Next lesson we will measure how likely each event is.'	Claim-Evidence-Reasoning with Consensus Discussion. Learners use their own listed outcomes as evidence for the claim that events are subsets, then justify with reasoning. Comparing predictions to evidence and revising them makes the shift in their ideas explicit, and applying the idea to Wanjiru's and Captain Otieno's different events shows why the same sample space supports different decisions.	Look for: (1) correct subset notation: $\{2, 4, 6\}$, {HH, HT, TH}, {rain, storm}, {storm}; (2) learners explain that an event is made of outcomes from S and that no outcome outside S can be in an event; (3) learners revise at least one prediction with a reason; (4) CER statements that reference actual lists. Use a 3-question exit check on mini whiteboards: 'Write S for one coin', 'Write the event "a number greater than 4" on a die' ($\{5, 6\}$), 'Is {rain} an event?' (yes). Count how many of the 5 cold-called learners answer correctly; follow up with those who are unsure.
Driving Question Board (DQB) Creation  VIDEO: Subsets of sample spaces Source: Khan	Learners revisit the Driving Question Board: 'How is probability applied in real life situations?' In groups they write on sticky notes (a) what they have now figured out and (b) new questions. Examples: 'Before a forecast is made, forecasters list all possible weather	Say: 'Look at the DQB. What can we now say about how a forecast like 70% is built?' WAIT TIME 10-15 seconds. Cold-call 3 groups to place one note and explain its heading. Ask: 'Which earlier question gets a small piece of an answer today?' Use probing: 'What	Driving Question Board Tracking. Moving and adding questions makes learners' growing understanding visible and keeps the phenomenon at the centre of the sequence. It also shows learners that today's learning (a complete list of outcomes) raises	Observable indicators: (1) at least one note per group states something learned using the terms sample space, outcome or event; (2) new questions ask for a method to turn outcomes into a number (for example 'How do we get 1/2 for even?'); (3) learners link

Academy (English - US curriculum) Search ARES for similar videos HTML: Probability of Compound Events (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	states: rain, storm, dry' (answer); 'How do we turn a list of outcomes into a number like 0.6?' (question); 'Are all outcomes equally likely, as with a coin, or not, as with rain?' (question); 'What does it mean when two events cannot happen together?' (question). They place notes under the headings 'How do we determine the probability of an event?' and 'How do we apply probability in real life situations?', then move one earlier question (for example 'Why does 70% not guarantee rain?') and mark it as partially answered.	do we still not know?' Point out that the sample space {rain, storm, dry} and the Kericho record of 14 rainy days out of 20 are connected: the record tells us what happened, the sample space tells us what could happen. Sort questions into 'next lesson' (theoretical probability, favourable / total outcomes) and 'later lessons' (combining events). Do not answer the questions yet.	the need for the next step (measuring likelihood).	notes to Wanjiru, Captain Otieno or the county office; (4) learners can say which DQB question was partly answered and why.
Model Building Phase VIDEO: Theoretical probability distribution example: tables <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Probability of Compound Events Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners add to the class model of the forecast (started in Lessons 1-2). In groups they draw a poster-style diagram with three parts: (1) a box labelled 'Sample space $S = \{\text{rain, storm, dry}\}$' showing all possible day states; (2) inside it, an oval for the event 'rain as the farmer sees it' = {rain, storm} and a smaller oval for 'storm' = {storm}; (3) a side panel showing the coin and die sample spaces as simpler models of the same idea (Heads = rain, Tails = no rain; even = {2, 4, 6}). They add an arrow labelled 'Next: how likely is each outcome?' and a sentence: 'A forecast is about an event, a subset of all possible outcomes.' They then list one thing the model explains and one thing it cannot yet explain (why 70% and not another number; whether a 70% day can stay dry).	Say: 'Add to our model. What is new since Lesson 2?' Hand out chart paper and markers. Prompt: 'Where does Wanjiru's event sit in the box? Where does Captain Otieno's event sit? What is inside both?' Since {storm} is inside {rain, storm}, ask: 'What does this tell you about the two events?' WAIT TIME 10-15 seconds. Cold-call 3 groups for a gallery-walk explanation (30 seconds each). Ask: 'What can our model not yet explain?' Record gaps on the board for Lesson 4. Remind learners not to write numbers beyond those learned; the established values (0.7, 0.3, 0.1, 0.6) are used in later lessons. Close with: 'Today we learned what could happen. Next we ask how likely each outcome is.'	Cumulative Model Revision with Gallery Walk. Learners extend the earlier model of repeated trials by adding a structure (sample space and events) that tells what the trials are choosing from. Showing the model's limits creates a purposeful link to Lesson 4.	Check posters for: (1) a correctly drawn S containing rain, storm and dry; (2) events drawn as parts of S (subsets) and not as new outcomes; (3) both coin and die examples linked to the weather model; (4) a stated limitation of the model. Use a 4-point checklist (S correct, event correct, link to phenomenon, limitation named) and record which groups need support before Lesson 4.

D. TEACHER REFLECTION

1. Which learners wrote HT and TH as one outcome at the Predict Phase, and did the two-coin experiment change their thinking? 2. Could learners list a complete

sample space systematically, or did they list only outcomes they saw in the trials? 3. Did learners understand that an event is a subset (for example {rain, storm}) and that different people can define different events from the same S? 4. How well did learners link today's lists to Wanjiru, Captain Otieno and the county office? 5. Which DQB questions should open Lesson 4 (theoretical probability = favourable / total outcomes)? 6. What adjustments to time or grouping are needed, especially for Station D and the poster?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What we observed: In our coin, two-coin and die activities, we saw these outcomes: one coin {H, T}; two coins {HH, HT, TH, TT} (HT and TH are different); die {1, 2, 3, 4, 5, 6}. A weather day on the KMD card has three states: rain, storm, dry.
What did I learn?	What we learned: A sample space S is the complete list of all possible outcomes of a random experiment. An event is a subset of S, for example 'even number' = {2, 4, 6}, and 'rain, as the farmer sees it' = {rain, storm}. An event can have one outcome, several outcomes or all outcomes of S.
How does this explain the phenomenon?	How this helps explain the phenomenon: Before KMD can say '70% chance of rain', it must list all possible day states S = {rain, storm, dry} and decide which outcomes make up the event. Wanjiru, Captain Otieno and Ms Achieng each care about a different event within the same sample space. We still need to find how likely each outcome is (Lesson 4).

LESSON 4 (80 minutes (2 x 40-minute lessons)): Probability Space II: Theoretical Probability and Why It Differs from What We Observe

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners calculate theoretical probability as favourable outcomes divided by total equally likely outcomes, compare it with experimental results, and explain why the two differ, so they can revise the forecast model and see where the 70% rain forecast numbers come from.
Knowledge	Theoretical probability = number of favourable outcomes / total number of equally likely outcomes. For a fair coin $P(H) = 1/2 = 0.5$; for a fair die $P(\text{even}) = 3/6 = 1/2$. The day-state probabilities $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$ and $P(\text{dry}) = 0.3$ are not equally likely, so they cannot be found by counting outcomes; they sum to 1.0. Experimental probability (class pooled $42/80 = 0.525$) is close to but not exactly the theoretical value of 0.5, and the difference is due to chance variation in a limited number of trials.
Skills	Calculating probabilities as fractions, decimals and percentages; comparing experimental and theoretical values; calculating the difference between them; explaining differences using evidence; checking that probabilities sum to 1; revising a model. NGSS practices: Analysing and Interpreting Data; Using Mathematics and Computational Thinking; Developing and Using Models; Constructing Explanations.
Attitudes	Curiosity about why numbers that do not guarantee an outcome can still be trusted; willingness to revise ideas when evidence shows a gap; respect for others' reasoning during group discussion; honesty in reporting data as collected.
Key Inquiry Question	1. How do we determine the probability of an event? 2. How do we apply probability in real life situations?
Purpose in Storyline	Lessons 1-2 gave us experimental probability (relative frequency, e.g. $42/80 = 0.525$ and Kericho $14/20 = 0.7$). Lesson 3 gave us the sample space and events as subsets. Lesson 4 joins the two: learners calculate theoretical probability from the sample space, compare it with experimental results and explain the gap. They then revise the forecast model by recognising that the KMD figure of 0.7 comes from records and models, because the day states are not equally likely. This prepares for the laws of probability in Lessons 5-8 and the decisions in Lesson 9.
Safety Notes	Use coins, bottle tops and dice on desks or mats, not on the floor, to avoid slipping. Do not throw dice across the room. Keep small objects away from mouths. Groups of 5 should share roles (tossers, recorder, checker, reporter, materials manager) so everyone is involved.

B. LESSON OVERVIEW

Mr Kamau reopens the puzzle: Why do Wanjiru Chebet near Eldoret, Captain Otieno at JKIA and Ms Achieng's Nairobi County office act on a 70% forecast that does not guarantee rain? Learners recall their pooled coin result of 42 heads in 80 tosses (0.525) and ask why it was not exactly 0.5. They predict, then use the sample space $\{H, T\}$ to calculate the theoretical $P(H) = 1/2$, and use a fair die to calculate $P(\text{even}) = 3/6 = 1/2$. Next they run a short die experiment and compare it with theory, explaining differences using chance variation and the number of trials. They then test whether the equally likely counting method works for the weather sample space $S = \{\text{rain, storm, dry}\}$. It does not, because the states are not equally likely, so the probabilities 0.6, 0.1 and 0.3 must come from records and models. Learners check that these sum to 1.0, and they revise the class forecast model and Driving Question Board (DQB).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Experimental versus theoretical probability simulation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12  Search ARES for similar readings	Learners look at the class pooled coin result from Lessons 1-2 (42 heads and 38 tails in 80 tosses, relative frequency $42/80 = 0.525$) and the Kericho record (14 rainy days in 20, $14/20 = 0.7$). Individually, in 3 minutes, they write predictions for: (a) What should $P(H)$ be for a fair coin if we could count the outcomes in the sample space $\{H, T\}$? (b) If we roll a fair die, what is $P(\text{even number})$? (c) Will our own die results match that number exactly? Why or why not? Then they discuss with a partner and sketch a quick prediction on a mini-whiteboard: a number line from 0 to 1 showing where they think $P(H)$ and $P(\text{even})$ lie.	Display the data on the board: 'Class pooled: 42 heads out of 80 = 0.525.' Say: 'Last week our coin gave 0.525. Wanjiru Chebet has to decide on a 0.7 chance of rain. Before I teach anything, what number would you expect for heads on a fair coin, and why?' Give 3 minutes of silent writing. Then say: 'Think about whether our results should match it exactly.' WAIT TIME 10-15 seconds after each question before taking answers. Cold-call 4 learners with different predictions (include at least 1 who predicts exactly 0.5, 1 who predicts it will not be exact, and 1 who predicts a value for the die). Record each prediction on the board without judging, writing 'Prediction' next to each. Say: 'Interesting. Hold on to these; we will test them today.'	Prediction with reasoning (Predict-Observe-Explain): learners commit to an idea and a reason before seeing evidence, so the later gap between prediction and result becomes something they need to explain. Linking the coin data to the forecast shows why this matters for Wanjiru's decision.	Check the mini-whiteboards: Do learners write a prediction with a reason rather than only a number? Look for at least 70% of learners naming $1/2$ or 0.5 for $P(H)$ and giving a reason, such as 'two sides'. Listen for misconceptions such as 'if 6 heads came, the next must be tails' or 'results must match theory exactly'. Note names for follow-up in the Explain phase.
Observe Phase  VIDEO: Experimental versus theoretical probability simulation Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Theoretical and Experimental Probability Practice Source: CK-12  Search ARES for similar readings	Part A (15 min): Learners use the sample space from Lesson 3 to build the theoretical idea. In groups of 5 they complete a 'Theory Card': (1) Coin: $S = \{H, T\}$; favourable for heads = 1; total = 2; $P(H) = 1/2 = 0.5 = 50\%$. (2) Fair die: $S = \{1, 2, 3, 4, 5, 6\}$; event 'even' = $\{2, 4, 6\}$; $P(\text{even}) = 3/6 = 1/2$. They also calculate $P(5) = 1/6$ and $P(\text{number less than 3}) = 2/6 = 1/3$. Part B (15 min): Each group rolls a die 30 times (or tosses a coin 30 times if dice are scarce) and records the number of evens on a tally table, calculates relative frequency of evens, then writes the difference from 0.5 (e.g. a group gets 17 evens: $17/30 = 0.57$, difference 0.07). The class then pools the results of 8 groups (240 rolls) on the board and calculates the pooled relative frequency. Part C (10 min): In pairs, learners	Part A: Model one calculation, then release: 'Favourable outcomes on top, total equally likely outcomes on the bottom. Which outcomes make heads? How many outcomes in total?' Circulate and ask: 'Why do we write $3/6$ and not $3/3$?' Part B: Hand out dice and tally sheets, assign roles, and say: 'Predict first: will your group get exactly half evens? Write your guess at the top of the sheet.' While groups roll, ask 3 groups: 'What is your relative frequency so far, and how far is it from 0.5?' Pool the results on the board and ask: 'Is the pooled result closer to 0.5 than most single groups were?' WAIT TIME 10-15 seconds. Cold-call 3 learners to report. Part C: Ask: 'Our sample space has 3 outcomes. Does that mean each has chance $1/3$? Look at the Kericho record of 14 rainy days in	Data comparison and contrast (Analysing and Interpreting Data): learners place theory (calculated from the sample space) beside experiment (their own tallies) and the weather values. Seeing both side by side, and seeing where counting fails for the weather states, builds the idea that the two kinds of probability answer the same question in different ways.	Observe whether groups: (a) write favourable/total correctly as $1/2$ and $3/6$ with simplification; (b) compute relative frequency correctly from their tallies; (c) calculate the difference between experimental and theoretical values; (d) check that $0.6 + 0.1 + 0.3 = 1.0$. Look for the error of using $1/3$ for all weather states and note which pairs spot why it is wrong. Collect the tally sheets as evidence.

	revisit the weather sample space $S = \{\text{rain, storm, dry}\}$ from Lesson 3 and test the counting method: 'There are 3 outcomes, so $P(\text{rain}) = 1/3$?' They compare this with the given values $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$, $P(\text{dry}) = 0.3$ and check that $0.6 + 0.1 + 0.3 = 1.0$.	20 and the KMD figures. Which is more believable for a March day in Kericho?' WAIT TIME 10-15 seconds, then cold-call 3 pairs. Do not state the answer. Let learners notice that the outcomes are not equally likely, so counting cannot give 0.6, 0.1 and 0.3.		
Explain Phase  VIDEO: Theoretical probability distribution example: tables <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Theoretical and Experimental Probability (Flexbook) Source: CK-12  Search ARES for similar readings	Learners write a Claim-Evidence-Reasoning (CER) explanation in their groups: 'Why was our experimental result (e.g. $42/80 = 0.525$, or our group die result) not exactly equal to the theoretical probability?' Evidence must cite at least two numbers (e.g. 0.5 vs 0.525, a difference of 0.025; their die pooled result). Reasoning must include: each trial is random; small numbers of trials vary more; the results tend to get closer to theory as trials increase (from Lesson 2: running relative frequency 0.6, 0.55, 0.55, 0.525). Learners then answer in writing: 'Why can't we use favourable/total outcomes to get $P(\text{rain}) = 0.7$ for Kericho?' (Outcomes in the weather space are not equally likely, so the probability comes from records such as 14/20 and from models.) They finish with a 'Wanjiru check': 'If the true chance of rain is 0.7, can Wanjiru still have a dry day? Explain.'	Provide a CER frame on the board: 'Claim: The difference occurred because _____. Evidence: _____ vs _____. Reasoning: _____.' Say: 'Use numbers from our table, not just words.' Circulate and prompt: 'If we tossed 8 coins only, would you be surprised by 5 heads? What about 8000 coins and 5000 heads?' WAIT TIME 10-15 seconds before groups respond. Choose 3 groups to share, deliberately including one whose result was far from 0.5. Ask the class: 'Does a result far from 0.5 mean the coin or die is unfair? What other explanation is there?' Cold-call 2 learners to summarise. Address misconceptions directly: 'Theoretical probability does not predict one toss; it describes the long run.' Link to the phenomenon: 'This is why a 70% forecast does not guarantee rain on one day.'	Claim-Evidence-Reasoning (Constructing Explanations): learners must tie a conclusion to specific numbers and a general principle, which stops 'it was just luck' from being the whole answer. Whole-class comparison of group explanations then sharpens the shared idea that experimental values scatter around theoretical ones.	Success criteria: the claim names chance variation or too few trials; the evidence quotes two specific values and their difference; the reasoning links to Lesson 2's idea that relative frequency settles as trials increase. Look for learners who say a different result means an unfair coin without evidence, and prompt them to look at the size of the difference. In the Wanjiru check, look for 'yes, 30% of days are still dry in the long run' or similar.
Driving Question Board (DQB) Creation  VIDEO: Probability of a compound event <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English	Learners revisit the Driving Question Board with the question 'How is probability applied in real life situations?' Each group reviews existing questions and (1) places a green dot on any question now partly answered (e.g. 'Why does 70% not mean it will rain?', 'How are forecast numbers worked out?') with a sticky note stating the new evidence; (2)	Say: 'Go to the board. Which questions can we now answer or partly answer using today's evidence?' Allow 5 minutes. Ask 3 groups to read out a green-dot question and the evidence on their note. WAIT TIME 10-15 seconds before calling on groups. Then say: 'What new puzzles has today created?' Cold-call 3 learners to read new questions. Group the	Question tracking with a Driving Question Board: marking progress and capturing new puzzles shows learners that science and mathematics advance by turning answered questions into new ones, and it keeps the anchoring phenomenon at the centre.	Look for sticky notes that cite evidence (a number or a calculation) rather than just opinions, and new questions that go beyond recall (e.g. asking about combining or deciding). Count how many groups produce at least one new question related to combining probabilities or decision-making. Note any groups whose notes show confusion

- US curriculum) Search ARES for similar videos EXERCISE: Theoretical and Experimental Probability Practice Source: CK-12 Search ARES for similar readings	writes at least one new question on a sticky note, e.g. 'If rain and dry cannot happen together, how do we add their chances?' or 'How do we combine two days of rain?' or 'How do 0.6, 0.1 and 0.3 get decided if we can't count outcomes?'. They sort the new notes into categories: how numbers are found, how numbers combine, how to decide.	new questions under headings and point out that the 'how numbers combine' questions will be answered in Lessons 5-8. Keep the Kericho 14/20 record and the forecast numbers visible beside the board.		about equally likely outcomes.
Model Building Phase VIDEO: Random numbers for experimental probability <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Develop probability distributions: Theoretical probabilities <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners revise their class 'Forecast Model' poster. The model now has three connected parts: (1) Sample space: $S = \{\text{rain, storm, dry}\}$; (2) Probabilities of states: $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$, $P(\text{dry}) = 0.3$, with the check $0.6 + 0.1 + 0.3 = 1.0$; 'rain' for the farmer means rain or storm = 0.7; (3) Where numbers come from: theoretical = favourable/total for equally likely outcomes (fair coin, fair die); experimental = relative frequency from records (Kericho 14/20 = 0.7) and models, for outcomes that are not equally likely. Groups add an 'Old idea / New idea' box, e.g. Old: 'every outcome has the same chance'; New: 'only equally likely outcomes can be counted; weather probabilities come from records'. Each learner finishes a 3-line exit ticket: one thing the model explains better, one thing it still cannot explain, and what Wanjiru, Captain Otieno or Ms Achieng might do with these numbers.	Say: 'Our model must now show where each number comes from. Use the Lesson 3 sample space and today's calculations.' Give groups 8 minutes to update the poster. Circulate and ask: 'Where on your model does 0.5 appear? Where does 0.7 appear? Do they come from the same method?' WAIT TIME 10-15 seconds. Ask 3 groups to present one change; cold-call 2 listeners to say whether they agree and why. Emphasise that the numbers 0.7, 0.3, 0.1 and 0.6 stay fixed. Keep the focus on checking they sum to 1.0. Collect exit tickets. Preview: 'Next lesson we ask which events can happen together and which cannot, and whether one day's weather affects the next.'	Cumulative model revision (Developing and Using Models): updating one shared model with an Old idea / New idea box makes the change in thinking visible and connects theoretical and experimental probability to the forecast phenomenon.	Check the model poster for: sample space correctly written; three state probabilities correct and summing to 1.0; a clear statement of where theoretical and experimental values come from. Exit ticket indicators: names the limit that counting only works for equally likely outcomes; refers to specific numbers; links to a Kenyan decision-maker. Sort tickets into 'secure', 'nearly' and 'needs support' for the next lesson.

D. TEACHER REFLECTION

After the lesson, consider: Which learners expected experimental results to equal theory exactly, and did their thinking change after seeing the pooled data? Could learners explain why 1/3 is wrong for each weather state? Did any group say that a result far from 0.5 meant an unfair die, and how did you respond? Which questions on the DQB showed new thinking? What would you change in how you linked the coin and die work back to Wanjiru's decision? Note which learners need more practice

with favourable/total and with checking that probabilities sum to 1 before Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that our class pooled coin result ($42/80 = 0.525$) and our die results were close to, but not exactly equal to, the theoretical values of 0.5 for a fair coin and $1/2$ for $P(\text{even})$ on a die. We also saw that the weather states cannot be counted as equally likely, since $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$ and $P(\text{dry}) = 0.3$ add up to 1.0.
What did I learn?	We learned that theoretical probability = favourable outcomes / total equally likely outcomes, and that experimental probability = relative frequency from trials or records. The two usually differ a little because of chance variation, and the difference tends to shrink as the number of trials grows.
How does this explain the phenomenon?	A forecast of 70% rain does not guarantee rain because probabilities describe the long run, not one day. For weather, the numbers come from records such as Kericho's 14 rainy days out of 20 and from models, because rain, storm and dry are not equally likely, so Wanjiru, Captain Otieno and Ms Achieng can trust the numbers even though a single day may turn out differently.

LESSON 5 (40 minutes (1 lesson)): Mutually Exclusive and Independent Events: Can Rain and Sunshine Happen Together?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners classify pairs of events from weather and everyday life as mutually exclusive, independent or neither, so that in Lessons 6 and 7 they can choose the correct law to combine probabilities for Wanjiru, Captain Otieno and the Nairobi County disaster office.
Knowledge	By the end of the lesson, learners should be able to: (1) state that two events are mutually exclusive if they cannot happen together, so $P(A \text{ and } B) = 0$; (2) state that two events are independent if one does not affect the chance of the other; (3) use the overlap or non-overlap of events as subsets of the sample space to decide whether events are mutually exclusive; (4) classify pairs such as rain and dry on the same day, rain on Monday and rain on Tuesday, and die and coin events as mutually exclusive, independent or neither.
Skills	Classifying, reasoning with evidence, communicating reasoning, using sample space lists $S = \{\text{rain, storm, dry}\}$ and $\{\text{HH, HT, TH, TT}\}$ to test events, asking questions, and constructing explanations (NGSS SEP: Analyzing and Interpreting Data; Constructing Explanations; Engaging in Argument from Evidence).
Attitudes	Learners appreciate that careful reasoning about events prevents wrong decisions, show patience when a classification is challenged by a peer, and respect the contributions of others in group discussion.
Key Inquiry Question	KICD Key Inquiry Question 1: How do we determine the probability of an event? KICD Key Inquiry Question 2: How do we apply probability in real life situations? Lesson focus: How can we tell whether two events can happen together, and whether one changes the chance of the other?
Purpose in Storyline	Lessons 1-4 gave learners experimental and theoretical probability, sample spaces and events as subsets. Lesson 3 showed that events are subsets of S. Lesson 5 uses that idea to sort pairs of events into mutually exclusive, independent or neither. This sorting is the evidence learners need before Lesson 6 (addition law) and Lesson 7 (multiplication law). It advances the driving question by showing that before probabilities can be combined, we must know how the events relate. This lesson does NOT teach the addition or multiplication laws; those come next.
Safety Notes	Use coins, bottle tops and dice on desks or on a mat to avoid objects rolling on the floor. Learners should not throw objects across the room. Keep walkways clear during group movement for the gallery sort.

B. LESSON OVERVIEW

Mr Kamau returns to the 70% forecast for Kericho and the Rift Valley. Wanjiru Chebet near Eldoret, Captain Otieno at JKIA and Ms Achieng's Nairobi County office each look at more than one event, such as rain and dry, or rain on two days. Learners first predict whether pairs of events can happen together and whether one affects the other. They then gather evidence by testing pairs of events against sample spaces: $S = \{\text{rain, storm, dry}\}$ for one day, a two-coin space $\{\text{HH, HT, TH, TT}\}$, and a die and coin. They use the Kericho record (14 rainy days out of 20, relative frequency 0.7) to see that rain on one year's March day does not change the chance in another year. In the Explain phase they build a classification rule with two questions: Can they happen together? Does one change the chance of the other? They add entries to the DQB and revise the class model by adding a step: classify the events first. The lesson ends with a summary table to be used in Lessons 6 and 7.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Compound probability of independent events Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Independent, Conditional, and Mutually Exclusive Events (Flexbook) Source: CK-12  Search ARES for similar readings	(8 minutes) Learners sit in groups of 5. Mr Kamau displays the forecast: $P(\text{rain}) = 0.7$, $P(\text{storm}) = 0.1$, $P(\text{dry}) = 0.3$ for Kericho. He shows four pair cards: (a) rain and dry on the same day in Kericho, (b) rain on Monday and rain on Tuesday, (c) rain and storm on the same day, (d) the coin shows Heads and the die shows 6. On a Predict sheet, each learner writes for each pair: 'Can both happen together? Yes/No' and 'Does the first change the chance of the second? Yes/No/Not sure', with a one-line reason. Learners then share in groups and record the group's majority prediction and any disagreement. Wanjiru, Captain Otieno and Ms Achieng are named as the people who need to combine these events.	Say: "Wanjiru is not only asking 'Will it rain?' She is asking about two things at once. Before we calculate anything, we must understand how the events relate. Write your own prediction first, silently." Give 3 minutes of silent writing. Circulate and note 3 learners with 'Not sure' for later cold-calling. Say: "Share in your group. You must give a reason, not just 'Yes' or 'No'." Cold-call 4 groups, one pair each, and write predictions on the board without correcting: "Group 2, what did you predict for rain and dry on the same day, and why?" WAIT TIME 10-15 seconds after each question before calling on a learner. Say: "I am not saying right or wrong yet. We will test these with evidence."	Predict-Observe-Explain (POE) with Public Commitment: learners commit to written predictions and reasons before evidence. This surfaces the common ideas, such as 'if two events are unrelated, they cannot happen together', that will be tested in the Observe phase, and gives learners a reason to look closely at the evidence.	Look for: every learner wrote a Yes/No and a reason for all four pairs; at least 3 groups record a disagreement; some learners suggest rain and dry cannot happen together and justify with 'the day is either wet or dry'. Note learners who confuse 'cannot happen together' with 'do not affect each other'. Record these in your notes as the misconception to address in the Observe and Explain phases.
Observe Phase  VIDEO: Compound probability of independent events - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Mutually Exclusive Events (Flexbook) Source: CK-12  Search ARES for similar readings	(14 minutes) Learners gather evidence in three stations rotated by groups (4 minutes each, plus 2 minutes of transition). Station A, One-day weather: using the sample space $S = \{\text{rain, storm, dry}\}$ with $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$, $P(\text{dry}) = 0.3$, learners list the outcomes in each event for 'rain or storm (rain for the farmer)', 'dry', and 'storm'. They shade or write the subsets and check whether any outcome appears in both rain and dry, and in both rain (rain or storm) and storm. They record: overlap? $P(A \text{ and } B)$? Station B, Two days: they use the Kericho record (14 rainy days out of 20, relative frequency 0.7) and the model that days are independent. They examine 'rain on Monday' and 'rain on Tuesday' using the two-day outcomes RR,	Say: "At each station, your evidence must come from the sample space or the data, not from what feels right." Circulate with a checklist. At Station A, ask: "Which outcomes belong to 'rain'? Which belong to 'dry'? Is any outcome in both lists?" WAIT TIME 10-15 seconds. At Station B, ask: "After a Heads on the first toss, how many of your pairs had Heads on the second? After a Tails on the first toss? What do you notice?" Remind learners that 10 pairs is a small sample and numbers will vary, and recall Lesson 2's idea that relative frequency becomes more stable with more trials. At Station C, ask: "If I tell you the die shows 4, does the chance of Heads change?" Then: "If I tell you the die shows an even number, does the chance of 'the die shows	Data Analysis with Sample Space Testing: learners use the same tool (sample space lists from Lesson 3) to test both properties, rather than relying on intuition. Comparing three contexts (weather in one day, weather over two days, die and coin) helps them see that 'cannot happen together' and 'does not change the chance' are two different questions.	Look for: learners correctly identify no overlap between rain only and dry, and between storm and dry; learners observe that the second toss shows Heads about equally often whatever the first toss; learners write evidence from the sample space or tally (not just opinions); learners notice that 'die even' and 'die shows 2' do overlap and are linked. Note groups that fill in only the Yes/No column with no evidence; prompt them with 'Which outcomes show that?'

	RN, NR, NN, and compare: out of the Monday-rain outcomes (RR, RN), how likely is rain on Tuesday? Learners use the tossing of a coin twice (Heads = rain, Tails = no rain) and 10 pairs of tosses to count how often the second toss is Heads after the first is Heads and after the first is Tails, and observe that both are near the same value. Station C, Die and coin: learners list the 12 outcomes of one die and one coin {1H, 1T, ..., 6H, 6T} and examine 'die shows even' and 'coin shows Heads', checking overlap and whether knowing the die result changes the chance of Heads. They also examine the neither case: 'die shows even' and 'die shows 2' (they can happen together, and knowing one changes the chance of the other). At each station, learners fill an evidence table with columns: Pair of events Can they happen together? Does one change the chance of the other? Evidence.	2' change?" Cold-call 3 learners at the end of the rotation: "Read one row from your evidence table and defend it." Do not yet give the formal terms for the decision rule; let learners describe in their own words.		
Explain Phase  VIDEO: Independent & dependent probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mutually Exclusive Events Practice Source: CK-12  Search ARES for similar readings	(10 minutes) Learners return to their groups and use their evidence tables to build a classification rule. Step 1: Each group finalises definitions in their own words: mutually exclusive (cannot happen together, no overlap, $P(A \text{ and } B) = 0$) and independent (one does not change the chance of the other). Mr Kamau then confirms the vocabulary. Step 2: Groups classify eight pairs on cards using the two questions: (1) rain and dry on the same day in Kericho; (2) rain on Monday and rain on Tuesday; (3) storm and dry on the same day; (4) coin shows Heads and die shows 6; (5) die shows	Say: "Use the two questions every time: Can they happen together? Does one change the chance of the other?" Be explicit that these are separate questions. Warn learners about a common error: "Some people say two events that are mutually exclusive are also independent. Think about rain and dry: if it rains, can it be dry? So does knowing rain change the chance of dry?" WAIT TIME 10-15 seconds. Cold-call 4 learners during the gallery: "Tell me one card you challenged and what evidence you used." Cold-call 2 more to explain how their prediction changed. Stress that the number 0.7 is not changed and	Claim-Evidence-Reasoning (CER) with Peer Challenge: each classification is a claim supported by evidence from the sample space or data, and a reason linking evidence to the definition. Peer challenges push learners to test their reasoning and separate the two properties. Revisiting the Predict sheet makes the change in thinking visible.	Look for: pairs 1, 3, 5 and 7 classified as mutually exclusive with reason 'no overlap in the sample space'; pairs 2 and 4 classified as independent with reason 'one does not change the chance of the other'; pair 6 classified as neither, with the reason that they can happen together and are linked; pair 8 classified as neither. Learners correct the idea that mutually exclusive events are independent. Exit check: ask each learner to write one pair of events for the farmer, pilot or county office and classify it with a reason.

	even and die shows odd; (6) die shows even and die shows 2; (7) rain only and storm on the same day; (8) Captain Otieno's flight is delayed and storm on the route day (neither, as they are linked and may not be exclusive). Learners place each card on a three-column board: Mutually exclusive Independent Neither, and write one sentence of reasoning. Step 3: Gallery discussion: groups visit another group's board and use sticky notes to challenge one card, writing 'Why?' Step 4: Learners revisit their Predict sheet, mark each prediction as supported or changed, and write a short explanation of what changed their thinking. Learners also write why this classification matters for Wanjiru, Captain Otieno and Ms Achieng.	that this lesson only classifies; the laws come in Lessons 6 and 7. Close with: "Why does Wanjiru need to classify first before combining probabilities?" Take 2 responses.		
Driving Question Board (DQB) Creation  VIDEO: Mutually Exclusive Events Principles Source: CK-12  Search ARES for similar videos  EXERCISE: Dependent and independent events Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(4 minutes) Learners return to the Driving Question Board with the driving question 'How is probability applied in real life situations?' and the KICD key inquiry questions. In groups they write 1-2 sticky notes: (a) what this lesson has answered, for example 'Rain and dry on the same day cannot happen together, so they are mutually exclusive' or 'Rain on Monday and rain on Tuesday do not change each other, so they are independent'; and (b) new questions, for example 'How do we add the probabilities of mutually exclusive events like storm and dry?' or 'How do we find the chance of rain on both Monday and Tuesday?' Learners place notes beside earlier questions on forecasts, relative frequency and sample spaces, and link related notes with	Say: "Look at the board. Which of our earlier questions got closer to an answer today? What new question do you have about combining the chances for Wanjiru or Ms Achieng?" WAIT TIME 10-15 seconds before taking ideas. Cold-call 3 groups to read a note and place it. Select 2 new questions about combining probabilities and mark them with a star: "We will answer these in Lessons 6 and 7." Ask: "Which of our DQB questions still need evidence?"	Question Clustering on the DQB: learners connect new questions to earlier ones and see how the sequence of lessons is building toward the driving question. Having learners link answered and open questions makes progress visible and keeps motivation toward the final decision task.	Look for: sticky notes that use the terms mutually exclusive and independent correctly; at least one new question about adding or multiplying probabilities; placement of notes next to relevant earlier questions; learners who can state which questions remain open. Note learners whose questions repeat earlier lessons without moving forward.

	string or arrows.			
Model Building Phase  VIDEO: Compound probability of independent events Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Independent, Conditional, and Mutually Exclusive Events (Flexbook) Source: CK-12  Search ARES for similar readings	(4 minutes) Learners open their class model of the forecast, which already includes relative frequency (Lessons 1-2), the sample space and events (Lesson 3) and theoretical probability, with day-state probabilities 0.6, 0.1, 0.3 (Lesson 4). They add a new step: 'Classify the events first: can they happen together? does one change the chance of the other?' Using arrows, they link 'mutually exclusive' to a note 'rain only and storm and dry: no overlap' and 'independent' to 'rain Monday and rain Tuesday', leaving boxes 'to be completed' for addition law and multiplication law. They update a summary table for the three decision makers: Wanjiru (rain on two days: independent), Captain Otieno (storm and dry on the route day: mutually exclusive), and the county office (rain only, storm and dry: mutually exclusive). Learners write one sentence: 'Before combining probabilities, I must decide whether the events are ____ or ____.'	Say: "Our model now has a new step. Where does it fit between listing the outcomes and calculating?" WAIT TIME 10-15 seconds. Cold-call 3 learners to explain one arrow. Ask: "Which decision maker uses mutually exclusive events? Which uses independent events?" Remind learners that the numbers 0.7, 0.3, 0.1 and 0.6 are unchanged and that the combining rules will come next. Collect the exit sentences and the summary table prompts to sort learners who need additional support before Lesson 6.	Cumulative Model Revision: learners add the new idea to a shared model, showing how classifying events fits between the sample space and the laws of probability. Linking each classification to Wanjiru, Captain Otieno and Ms Achieng keeps the model connected to the anchoring phenomenon.	Look for: model has a clear 'classify first' step; arrows connect mutually exclusive to non-overlapping events and independent to events that do not change each other's chance; learners match the right type of events to each decision maker; exit sentences correctly state that the classification decides which law to use. Note learners who still treat the two terms as the same.

D. TEACHER REFLECTION

Which learners still confused 'cannot happen together' with 'does not affect each other'? Which evidence (sample space lists, coin pairs or the die and coin) was most convincing to them? Did the 10 pairs of coin tosses produce results that confused learners because of small samples, and how did you link this to Lesson 2? Which learners are ready to use these classifications in the addition law (Lesson 6) and the multiplication law (Lesson 7), and who needs more practice with the two questions? Did learners connect the classifications to Wanjiru, Captain Otieno and Ms Achieng, or did they treat the task as abstract cards?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did you observe? For rain and dry on the same day, did any outcome appear in both events? For rain on Monday and rain on Tuesday (and for the first and second coin toss), did the chance of the second change after the first? For die even and coin Heads, and for die even and die shows 2, what did you find?
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What did I learn?	What did you learn? Mutually exclusive events cannot happen together, so $P(A \text{ and } B) = 0$. Independent events do not change each other's chance. Some pairs are neither. These are two different questions.
How does this explain the phenomenon?	How does this help explain the 70% forecast puzzle? Explain why Wanjiru, Captain Otieno and Ms Achieng must first decide whether events are mutually exclusive, independent or neither before combining probabilities such as $P(\text{rain only}) = 0.6$, $P(\text{storm}) = 0.1$ and $P(\text{dry}) = 0.3$, or rain on two days.

LESSON 6 (40 minutes (one double or two single periods may be used; adjust to timetable)): Laws of Probability I: The Addition Law and the Complement Rule

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the addition law $P(A \text{ or } B) = P(A) + P(B)$ for mutually exclusive events and the complement rule $P(\text{not } A) = 1 - P(A)$ to the KMD day-state probabilities, so they can work out the chance of rain or storm for farmer Wanjiru's decision.
Knowledge	Learners will state and apply the addition law for mutually exclusive events; state and apply the complement rule $P(\text{not } A) = 1 - P(A)$; explain why the addition law only works when events cannot happen together ($P(A \text{ and } B) = 0$); and show that 'rain' for the farmer means rain only or storm = $0.6 + 0.1 = 0.7$.
Skills	Using mathematics and computational thinking (NGSS SEP): calculating probabilities with the laws; analysing and interpreting data (KMD day-state table); constructing explanations; checking that probabilities of all outcomes sum to 1; communicating reasoning with fractions, decimals and percentages.
Attitudes	Curiosity about how forecasts are built from numbers; precision and honesty when checking calculations; respect for peers' reasoning in group talk; appreciation that mathematics supports real decisions made by farmers like Wanjiru, pilots like Captain Otieno and county offices.
Key Inquiry Question	1. How do we determine the probability of an event? 2. How do we apply probability in real life situations?
Purpose in Storyline	Lessons 4-5 gave learners the day-state probabilities (rain only 0.6, storm 0.1, dry 0.3, summing to 1) and the idea of mutually exclusive events. Lesson 6 gives the first law for combining those probabilities, which explains how the headline 70% forecast is made up and how the 30% 'no rain' figure follows from it. Lesson 7 will add the multiplication law for independent events.
Safety Notes	Low-risk classroom lesson. Keep bags and chairs out of walkways during group movement to the DQB. If using bottle tops or counters for the optional card-sort, ensure small items stay on desks and are not put in mouths. No outdoor activity required.

B. LESSON OVERVIEW

Mr Kamau opens by revisiting the Driving Question Board and the evidence so far: relative frequency (Kericho 14/20 = 0.7), the day-state sample space $S = \{\text{rain, storm, dry}\}$ with probabilities 0.6, 0.1 and 0.3, and the idea of mutually exclusive events from Lesson 5. Learners predict how the KMD forecast of 70% rain can be linked to the separate day-state numbers. In the Observe phase, groups of 5 use a 10 x 10 grid of 100 'similar March days' (60 shaded rain only, 10 storm, 30 dry) to count and discover that combining mutually exclusive outcomes means adding their probabilities, and that 'not rain' is what is left over after 1 is shared out. In the Explain phase, learners state the addition law and complement rule in their own words, test where the law fails (events that can happen together), and apply both to Wanjiru's planting decision, Captain Otieno's storm check and the county office's flood preparation. They then add new questions to the DQB and revise the class model of the forecast to include the two laws.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Addition rule for probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Complement Rule for Probability (Flexbook) Source: CK-12  Search ARES for similar readings	Learners look at the anchor forecast on the board: 'KMD: 70% chance of rain in Kericho and the Rift Valley; 10% chance of storm.' Wanjiru's problem is posed: 'She hears 70% for rain and 10% for storm. Does she add them, or is the storm already inside the 70%?' The day-state table from Lesson 4 is shown (rain only 0.6, storm 0.1, dry 0.3). Individually, learners write for 2 minutes: (a) their prediction for the chance of 'rain or storm' and (b) the chance of 'no rain', with a one-sentence reason. They then share with an elbow partner and note any disagreement.	Display the forecast and table. Say: 'Wanjiru near Eldoret must decide whether to plant today. Before we calculate anything, what do you think the chance of rain or storm is? Write your prediction and your reason.' Give 2 minutes of silent writing. Then say: 'Turn to your elbow partner and compare.' Circulate and listen for learners who add, who keep 0.7, or who say 0.6 only. Do NOT confirm any answer. Cold-call 3 learners with different predictions: 'Amina, what did you predict and why?' After each response, WAIT TIME of 10-15 seconds, then ask a different learner: 'Who can say that in a different way?' Record the 3 predictions on the board under 'Our predictions' and say: 'We will test these with evidence.'	Prediction with reasoning (Predict-Observe-Explain). Writing a prediction and a reason before instruction makes learners commit to their current ideas about combining probabilities, so the later evidence can confirm or challenge them. Comparing predictions in pairs surfaces the common misconception that probabilities of day states 'overlap' with the forecast.	Observe and note: how many learners write a numerical prediction (not just 'high'); how many give a reason that refers to the table; how many mention adding. Tally the three prediction types on the board (about 0.7, about 0.8 by adding $0.7 + 0.1$, about 0.6). Look for learners who use the Lesson 5 idea that rain and dry cannot happen together on the same day. Record names of learners who add $0.7 + 0.1$ so you can revisit their reasoning in the Explain phase.
Observe Phase  VIDEO: Addition rule for probability - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Complement Rule for Probability Practice Source: CK-12  Search ARES for similar readings	In groups of 5, learners receive a 10 x 10 grid representing 100 similar March days in Kericho (60 squares pre-shaded as 'rain only', 10 as 'storm', 30 as 'dry'). Tasks: (1) Count squares for each state and write each as a fraction of 100 and as a decimal. (2) Count the squares that are 'rain only or storm' and write $P(\text{rain or storm})$. Compare with $0.6 + 0.1$. (3) Count 'storm or dry' squares and compare with $0.1 + 0.3 = 0.4$. (4) Count the squares that are NOT rain (not rain only and not storm) and compare with $1 - 0.7 = 0.3$. (5) Complete a table with columns: Event, Squares, Probability, Sum of separate probabilities. (6) Try the event 'rain only and storm' and decide why no square can belong to both. Each learner has a role: counter, recorder, checker, reporter, materials manager.	Hand out grids and the recording table. Say: 'This grid is 100 days like our March morning. Your job is to find out how the separate chances join together. Count carefully, and check your partner's count.' Circulate to every group at least once. Prompt: 'Which squares did you count for rain or storm? Did any square get counted twice?' and 'What do you notice when you compare your count with the sum of the two separate probabilities?' If a group is stuck, ask: 'What happens if you count rain-only squares and then storm squares?' Do NOT state the law. After 12 minutes, ask each group's reporter to give one row of the table while you write it on the board. Cold-call 4 different reporters (one per row: rain or storm, storm or dry, not rain, rain only and storm). After each report, WAIT 10-15 seconds before	Pattern finding in data using a concrete model (area/grid model). Counting squares gives a visual, hands-on way to see that adding the probabilities of events that share no squares gives the probability of either happening. Comparing the count with the sum helps learners discover the addition law inductively. Role assignment ensures every learner contributes to the evidence.	Look for: correct counts (60, 10, 30; 70; 40; 30; 0); learners writing decimals that match the squares (0.6, 0.1, 0.3); groups noticing that 'sum of separate probabilities' equals the direct count; groups noticing that rain only and storm squares do not overlap. Listen for the phrase 'cannot be in both'. Use a checklist: counted correctly / compared with sum / noticed no overlap. Note any group that double counts the storm squares in 'rain' and plan a targeted question: 'Is a storm square also shaded as rain only?'

		inviting another group to agree or challenge: 'Does your group's count match? Show with a thumb.'		
Explain Phase  VIDEO: Mutually Exclusive Events Principles Source: CK-12  Search ARES for similar videos  HTML: Mutually Exclusive Events (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the board data and their tables. In their groups they write a class statement for the addition law: 'For events that cannot happen together, $P(A \text{ or } B) = \dots$ ' and for the complement: ' $P(\text{not } A) = \dots$ '. They apply the laws to three tasks: (1) Wanjiru: $P(\text{rain only or storm}) = 0.6 + 0.1 = 0.7$, and $P(\text{no rain}) = 1 - 0.7 = 0.3$. (2) Captain Otieno: $P(\text{no storm}) = 1 - 0.1 = 0.9$; $P(\text{storm or dry}) = 0.1 + 0.3 = 0.4$. (3) Ms Achieng's county office: which events can be joined by the addition law? A short 'Does it work?' sort: learners decide whether the addition law can be used for pairs such as rain only and dry; storm and dry; a die showing 2 and a die showing 5; and a pair that are NOT mutually exclusive (die 'even' and die 'greater than 3'). They explain why the law fails for the last pair, without calculating the general overlap rule. They then answer: 'Does the 70% forecast guarantee rain? What does 0.7 now mean in terms of the day states?'	Say: 'You have counted the evidence. Now write the rule in your own words. What must be true about the two events?' Give 3 minutes for group writing, then cold-call 3 groups to read their rule. After each, WAIT 10-15 seconds and ask: 'What would you change or add?' Build the class statements on the board: $P(A \text{ or } B) = P(A) + P(B)$ when A and B are mutually exclusive, $P(A \text{ and } B) = 0$; $P(\text{not } A) = 1 - P(A)$. Revisit the Predict phase: 'Who predicted 0.8? What did the grid show?' Invite a learner who added $0.7 + 0.1$ to explain what they now think. Ask: 'Why is the storm already inside the 70%?' For the sort task, say: 'Hold up green for the law works, red for it does not. Explain with the sample space.' Cold-call 4 learners with reasons. For the non-exclusive pair, say: 'Even and greater than 3 can both happen on 4 and 6. Because they overlap, simply adding would double count. We will not use the law there.' Close with: 'A 70% forecast does not promise rain; it tells us about about 7 days in 10. Why can Wanjiru, Captain Otieno and Ms Achieng still act on it?' Take 3 responses with WAIT TIME between each.	Constructing explanations and claim-evidence-reasoning, combined with a 'does the rule apply?' sorting task. Learners state a claim (the law), support it with their grid counts (evidence) and justify the condition 'mutually exclusive' (reasoning). Contrasting examples where the law works and fails sharpens understanding of when it can be used. Revisiting initial predictions lets learners see how their thinking changed.	Listen for: the condition 'cannot happen together' included in learner statements; correct calculations 0.7, 0.3, 0.9, 0.4; correct use of green/red cards (rain only and dry: green; storm and dry: green; die 2 and 5: green; even and greater than 3: red); a reason that refers to outcomes that appear in both events. Use a 3-question exit check on paper: (1) $P(\text{rain only or dry}) = ?$ (0.9) (2) $P(\text{not storm}) = ?$ (0.9) (3) Why can we add in question 1? Code responses as secure / developing / not yet. Plan to revisit with learners who give 'because they are different numbers' instead of 'they cannot both happen'.
Driving Question Board (DQB) Creation  VIDEO: Probability with Venn diagrams Source: Khan	Learners go to the DQB and add sticky notes in two colours: green for 'What we figured out today' (for example 'Rain only $0.6 + \text{storm } 0.1$ gives 0.7 for rain or storm' and 'No rain is $1 - 0.7 = 0.3$ ') and yellow for new questions (for example 'What is the chance of rain on two days	Say: 'Go to the board with one green note and one yellow note. Write one thing the evidence showed us and one new question we still cannot answer.' Give 4 minutes. Then cluster the notes with the class: 'Which questions belong together?' Cold-call 3	Driving Question Board as a tracking tool for growing understanding. Writing what was learned and what remains puzzling keeps learner questions at the centre, links today's evidence to the anchoring phenomenon, and shows how the unit's sequence of	Check that green notes state a correct law or result with numbers; that yellow notes are genuine questions linked to the forecast, the farmer, the pilot or the county office; that at least some questions anticipate multiplication of probabilities for two days. Note

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Mutually Exclusive Events Practice Source: CK-12 Search ARES for similar readings	in a row?', 'How do we combine chances when events do not exclude each other?', 'How would Wanjiru use these numbers across the whole week?'). Learners then link their notes to existing questions on the DQB, especially those about how forecast numbers are worked out and combined, and mark any older question that is now partly answered with a tick.	learners to read a question and say which lesson might help (for example the two-day question goes with Lesson 7 on the multiplication law). WAIT 10-15 seconds after each question before asking for a connection. Point to the driving question: 'How is probability applied in real life situations? What did the addition law add to our answer today?' Take 2 responses. Mark the questions answered today and keep the open ones visible for the next lesson.	lessons responds to learner thinking.	learners whose notes repeat Lesson 5 content, which may signal that they need more practice. Count notes per learner and look for quiet learners to invite in later lessons.
Model Building Phase VIDEO: Addition rule for probability Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Complement Rule for Probability (Flexbook) Source: CK-12 Search ARES for similar readings	Learners revise the class model of 'How the forecast works', drawn in groups on A3 paper or in exercise books. The model from earlier lessons includes past records (Kericho 14/20 = 0.7), the sample space $S = \{\text{rain, storm, dry}\}$ and probabilities 0.6, 0.1 and 0.3 summing to 1. Today they add: (1) a box showing the addition law $P(A \text{ or } B) = P(A) + P(B)$ with the condition 'mutually exclusive', (2) the working $0.6 + 0.1 = 0.7$ linking rain only and storm to the 70% forecast, (3) a box showing the complement $P(\text{no rain}) = 1 - 0.7 = 0.3$ and $P(\text{no storm}) = 0.9$, and (4) a short arrow note: 'Wanjiru can use the 0.7 and 0.3 to compare the cost of waiting with the cost of planting.' Groups also add one 'still to find out' arrow pointing to two-day weather. Each group writes a 2-sentence explanation answering: 'Why can a 70% forecast be trusted even though it is not certain?'	Say: 'Add today's evidence to your model. Show how the separate day states join up to give the forecast the farmer hears.' Circulate and ask: 'Where does the storm appear in your model? Is it inside the 70% or extra?' and 'Which law gives 0.3 and how do you know?' Remind groups to use the numbers 0.6, 0.1, 0.3 and 0.7 exactly. After 8 minutes, hold a gallery walk of 3 minutes, with each group leaving one reporter at the model. Cold-call 3 reporters to share one change they made. WAIT 10-15 seconds after each, then ask the class: 'What is the same or different in our models?' Close with a summary: 'Today we saw how probabilities of outcomes that cannot happen together are added, and how the chance of not happening is what is left from 1. Next lesson we will look at what happens across two days.' Hand out the summary table prompt for the exercise book.	Cumulative model building and revision. Updating a visual model with the new laws makes the structure of the forecast explicit (data, sample space, laws), links each lesson to the phenomenon and prepares learners to use the same model to justify decisions in Lesson 9. Gallery walk exposes learners to alternative representations and strengthens peer feedback.	Check models for: sample space and probabilities correctly shown; addition law with condition 'mutually exclusive'; correct working $0.6 + 0.1 = 0.7$ and $1 - 0.7 = 0.3$; the storm shown as part of the 0.7 rather than extra; a two-sentence explanation of trust in a non-certain forecast that mentions long run or many days. Use a quick rubric: complete / partly complete / needs support. Note any group whose model shows 0.8 or an altered number and give a prompt to re-check against the grid.

D. TEACHER REFLECTION

After the lesson, reflect: Which learners moved from adding $0.7 + 0.1$ to seeing that the storm is already inside the 70%? Did the 10 x 10 grid make 'mutually exclusive' visible, or did some learners still count squares twice? How well did learners explain why the law fails for overlapping events like 'even' and 'greater than 3' without needing the general overlap formula? Did the exit check show that learners can use $1 - P(A)$ correctly? Which questions on the DQB suggest readiness for the multiplication law in Lesson 7? What adjustments to group roles or time are needed (for example, shorten counting by pre-labelling rows)? Remember that the probabilities 0.6, 0.1, 0.3 and 0.7 must stay unchanged, and that the classroom thresholds are for learning and are not real KMD rules.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that on the grid of 100 similar March days in Kericho, 60 days were rain only, 10 were storm and 30 were dry; the days that were 'rain only or storm' numbered 70, the days that were 'storm or dry' numbered 40, and the days with no rain numbered 30. No day was counted in two states at once.
What did I learn?	We learned that for mutually exclusive events, $P(A \text{ or } B) = P(A) + P(B)$, and that $P(\text{not } A) = 1 - P(A)$. So $P(\text{rain only or storm}) = 0.6 + 0.1 = 0.7$, $P(\text{storm or dry}) = 0.1 + 0.3 = 0.4$, and $P(\text{no rain}) = 1 - 0.7 = 0.3$.
How does this explain the phenomenon?	This helps explain the forecast because the 70% chance of rain is made up of rain only (0.6) and storm (0.1), and the 30% chance of no rain is what remains. Farmer Wanjiru, Captain Otieno and the county office can use these numbers to compare the chances with their cost of being wrong, even though the forecast does not guarantee what will happen on one day.

LESSON 7 (40 minutes (one double lesson of 80 minutes may be used if available)): Laws of Probability II: The Multiplication Law and Two Rainy Days

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners use the multiplication law $P(A \text{ and } B) = P(A) \times P(B)$ for independent events to find the chance of rain on two consecutive days, so they can see how a single-day forecast of 0.7 is combined into a two-day probability that farmers, pilots and county officers can use.
Knowledge	Learners will state and apply the multiplication law for independent events: $P(A \text{ and } B) = P(A) \times P(B)$. They will calculate $P(\text{rain on both days}) = 0.7 \times 0.7 = 0.49$ and $P(\text{no rain on both days}) = 0.3 \times 0.3 = 0.09$, and find $P(\text{head and a 6})$ for a coin and a die.
Skills	Identifying whether two events are independent before multiplying (Lesson 5); using probability laws (Lesson 6); multiplying decimals and fractions accurately; using the NGSS practices of Using Mathematics and Computational Thinking, Developing and Using Models, and Constructing Explanations.
Attitudes	Curiosity about how numbers that do not guarantee outcomes can still guide decisions; respect for evidence from KMD and Kericho records; collaboration and perseverance in group work; honesty in checking one's own predictions against calculations.
Key Inquiry Question	1. How do we determine the probability of an event? 2. How do we apply probability in real life situations?
Purpose in Storyline	Lessons 1-4 showed where the 0.7 comes from (relative frequency 14/20 and a sample space of rain, storm, dry). Lesson 5 classified events as mutually exclusive or independent, and Lesson 6 used the addition law and the complement. This lesson gives learners the multiplication law so they can answer the question Wanjiru Chebet must ask: what is the chance it rains on both of the next two days? The values 0.49 and 0.09 feed straight into the tree diagram in Lesson 8 and the decisions in Lesson 9.
Safety Notes	Learners toss only a coin and roll a die at desks. Keep objects on the desk surface so they do not roll away or cause trips. No other special hazards. Remind learners to keep bags out of walkways during group work.

B. LESSON OVERVIEW

Mr Kamau opens by recalling Wanjiru Chebet, the maize farmer near Eldoret, who hears that the chance of rain is 70% today. She now wants to know the chance that it rains on BOTH of the next two days before she plants. Learners first predict the answer, then test the idea with coin and die experiments to see what happens when two independent events are combined. They find that multiplying probabilities fits the data, apply it to $P(RR) = 0.7 \times 0.7 = 0.49$ and $P(NN) = 0.3 \times 0.3 = 0.09$, and explain why the two-day chance is smaller than the one-day chance. They add the new ideas to the Driving Question Board and revise the class model of how forecast numbers are worked out and combined.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Learners sit in groups of 5. They are shown the situation: the KMD	Say: "Wanjiru has 2 acres of maize near Eldoret. KMD says 70%	Prediction with reasoning (Predict-Observe-Explain): committing to a	Look for: (a) learners who write a number as a decimal, fraction or

Compound probability of independent events - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Probability with general multiplication rule Source: Khan Academy (English - US curriculum) Search ARES for similar readings	forecast gives $P(\text{rain}) = 0.7$ for Kericho and the Rift Valley on each day, and Wanjiru Chebet wants to know the chance of rain on both of two consecutive days. Individually, in their notebooks, they write a prediction: Will the chance of rain on both days be greater than 0.7, equal to 0.7, or less than 0.7? They write one sentence of reasoning and a guess for the number. They then share in their groups and record the group's range of guesses on a sticky note.	chance of rain today and the same tomorrow. Before we calculate anything, what do you think is the chance that it rains on BOTH days? Greater than, equal to, or less than 0.7?" Give WAIT TIME of 10-15 seconds in silence before anyone writes. Walk around and read 6-8 notebooks without correcting. Cold-call 4 learners from different groups to give a prediction and reasoning: "Amina, what did you write and why?" Record all predictions on the board in three columns (greater, equal, less) with tallies. Say: "We are not marking these. We will return to them at the end."	written prediction and a reason exposes learners' existing ideas, such as 'two days of 70% rain make 140%' or 'it stays 70%'. Having to return to the prediction later creates a need to know and links to the DQB.	percent between 0 and 1 and not above 1; (b) the tally across columns, noting how many choose 'less than 0.7'; (c) reasoning that mentions both days or uses past data, such as 'rain on 14 of 20 days in Kericho'; (d) misconceptions such as adding $0.7 + 0.7 = 1.4$, which are noted for use in the Explain Phase.
Observe Phase VIDEO: Compound probability of independent events Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Probability of Independent Events (Flexbook) Source: CK-12 Search ARES for similar readings	Activity 1 (coin and die, about 12 minutes): In their groups, learners test whether multiplication works for independent events. Each group tosses a coin and rolls a die together 30 times (or splits 6 trials per learner), recording whether the coin shows Heads and the die shows a 6 (or an even number) on each trial. They record H and even in a tally table: heads only, even only, both. They calculate the relative frequency of 'Heads and even' and compare it with $P(H) \times P(\text{even}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = 0.25$. They also list the sample space of 12 outcomes (H1 ... T6) and count the favourable outcomes. Activity 2 (about 8 minutes): learners study the two-day weather sample space RR, RN, NR, NN. Using $P(R) = 0.7$ and $P(N) = 0.3$ they fill in a table: $P(RR) = 0.7 \times 0.7 = 0.49$ and $P(NN) = 0.3 \times 0.3 = 0.09$, and record that rain on Monday does not change the chance of rain on Tuesday (independent, from Lesson 5).	Distribute one coin, one die and a recording sheet per group. Say: "Roles: tosser, roller, recorder, checker, reporter. Rotate every 6 trials." Model one trial. Circulate and ask: "How many of your trials have BOTH a head and an even number? Is that more or less than half of the trials?" and "Does the coin land differently because of the die? Are these independent?" Give WAIT TIME of 10-15 seconds after each question. Then pool the results on the board: ask each group to report its 'both' count and add them up, so learners see that the pooled relative frequency is near 0.25. Cold-call 3 reporters: "What was your group's relative frequency? How does it compare to $\frac{1}{4}$?" For Activity 2, write on the board: "$P(R) = 0.7$ each day". Ask: "Does rain on Monday change the chance of rain on Tuesday?" and cold-call 2 learners to justify independence. Then ask groups to compute the four products and check that they add to 1.00 ($0.49 +$	Data-driven pattern finding with pooled class data: learners gather experimental evidence (relative frequency of heads and even), then compare it with the product $P(A) \times P(B)$. This uses the connection between experimental and theoretical probability from earlier lessons and shows the law as a pattern in data before it is stated as a rule. Using a sample space grid of 12 outcomes supports the idea that 'both' is 3 of 12 for some events, tying numbers to counting.	Observe: (a) learners tally correctly and compute relative frequency as number of 'both' trials divided by total trials; (b) groups' pooled results lie near 0.25 and learners comment on differences from 0.25 due to random variation; (c) learners multiply 0.7×0.7 correctly to get 0.49 and 0.3×0.3 to get 0.09 without adding; (d) learners state independence in their own words, such as 'yesterday's rain does not change today's chance'; (e) groups check that all four two-day products sum to 1.00.

		0.21 + 0.21 + 0.09).		
Explain Phase  VIDEO: General multiplication rule example: independent events Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Probability of Independent Events Practice Source: CK-12  Search ARES for similar readings	Learners return to the predictions from the Predict Phase and the board tallies. In groups they write a short explanation: 'The chance of rain on both days is ____ because ____.' They state the multiplication law in words and symbols: for independent events, $P(A \text{ and } B) = P(A) \times P(B)$. They explain why 0.49 is less than 0.7: the event 'rain on both days' is harder to achieve than 'rain on one day', so it is only about 49 in every 100 pairs of days. They also state how 'and' differs from 'or' (Lesson 6: addition law for mutually exclusive events) and why multiplication is used for independent events and addition for mutually exclusive ones. Finally, each learner solves a transfer problem: $P(\text{no rain on both days}) = 0.09$; and one new problem such as 'A fair die is rolled and a coin is tossed. Find $P(\text{a 6 and tails})$.'	Say: "Look at your first prediction. Was the chance of both days greater than, equal to, or less than 0.7? What does the data tell us?" Give WAIT TIME of 10-15 seconds. Cold-call 3 learners with different predictions to revise their thinking in front of the class: "What did you think before? What do you think now, and what evidence changed it?" Address the misconception of adding: write $0.7 + 0.7 = 1.4$ on the board and ask: "Can a probability be 1.4? What does that tell us?" Then ask: "How do we know when to multiply and when to add?" and have pairs talk for 1 minute before cold-calling 2 pairs. Write the law with a worked example: " $P(R \text{ and } R) = 0.7 \times 0.7 = 0.49$ means about 49 out of every 100 such pairs of days." Use Kenyan context: "If Kericho has 100 two-day periods like this, about how many have rain on both days?" Assign the transfer problem; circulate and check 8-10 notebooks. Do NOT use tree diagrams or the 'at least one' calculation yet; say "Next lesson we will organise all four outcomes in a tree diagram."	Claim-Evidence-Reasoning (CER): learners write a claim (the probability of rain on both days), cite evidence (0.7 each day from KMD and Kericho 14/20, and the coin and die class data), and give reasoning (the multiplication law for independent events). Revising earlier predictions encourages conceptual change and makes thinking visible.	Look for: (a) correct statement of $P(A \text{ and } B) = P(A) \times P(B)$ and the condition 'independent'; (b) 0.49 and 0.09 stated with the meaning 'about 49 in 100' and 'about 9 in 100'; (c) correct answer to the transfer problem: $P(6 \text{ and tails}) = 1/6 \times 1/2 = 1/12$; (d) learners distinguish when to add (mutually exclusive 'or') and multiply (independent 'and'); (e) a revised claim that explicitly uses evidence. Misconception check: any learner still adding $0.7 + 0.7$ should be followed up immediately.
Driving Question Board (DQB) Creation  VIDEO: Compound probability of independent events - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES	Learners revisit the class DQB with the driving question 'How is probability applied in real life situations?' In groups they use sticky notes to (a) add what they have now learned (for example, 'We can find the chance of rain on two days by multiplying: 0.49'), (b) mark earlier questions that have been answered, such as 'How do forecasters combine chances?', and (c) add new questions, such as 'What is the chance of rain on	Say: "Look at the DQB. Which questions can we now answer? Which new questions have appeared?" Give WAIT TIME of 10-15 seconds. Cold-call 4 groups, one at a time, to read a note and place it. Ask: "Which question do you think we need next to help Wanjiru decide?" Prompt: "What if she needs to know about only one rainy day, or at least one?" Say that these will be handled in Lesson 8 with tree diagrams, and	Driving Question Board: it makes the class's growing understanding visible and gives learners ownership of the next steps. Sorting questions into answered and unanswered shows how each lesson's evidence moves them closer to the driving question and gives purpose to the next lesson.	Observe: (a) learners link their notes to the phenomenon (Wanjiru, Captain Otieno, the county office) and not only to the coin; (b) at least one note per group shows a correct new idea (the multiplication law, 0.49 or 0.09); (c) new questions are about combined events such as 'only one day' and 'at least one day'; (d) learners can say which lesson answered which question.

for similar videos EXERCISE: Probability with general multiplication rule Source: Khan Academy (English - US curriculum) Search ARES for similar readings	only one of the two days?' and 'What is the chance of at least one rainy day?' Each group posts at least one answered question and one new question.	that the decision for Captain Otieno and Ms Achieng follows in Lesson 9. Group questions by theme (combining chances, storms, decisions) and note them for the next lesson. Keep it to about 5 minutes.		
Model Building Phase VIDEO: Compound probability of independent events Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Probability of Independent Events (Flexbook) Source: CK-12 Search ARES for similar readings	Learners update the class model, a poster or notebook diagram titled 'How the 70% forecast is worked out and combined'. The existing model has: relative frequency from records ($14/20 = 0.7$), sample space and theoretical probability, mutually exclusive events and the addition law. Groups add a new panel: 'Two independent days: $P(\text{rain on both}) = 0.7 \times 0.7 = 0.49$; $P(\text{no rain on both}) = 0.3 \times 0.3 = 0.09$. Multiply for independent events and 'and'. They draw a simple arrow from 'one-day probability' to 'two-day probability' and write one sentence on how this helps Wanjiru. Finally, each learner completes an exit ticket: 'Rain on two days in a row is less likely than rain on one day because ____.'	Say: "Take out the class model. What new piece do we need to add today so that Wanjiru can use the forecast for two days?" Give WAIT TIME of 10-15 seconds. Cold-call 3 groups to propose the new panel and the wording; refine it with the class. Ask: "What does the 0.49 tell Wanjiru? What does it NOT tell her?" and expect the answer that it does not guarantee rain on both days. Ask: "What would change if the days were not independent?" and take 2 responses without teaching conditional probability: say only that we assume independence in our classroom model. Collect exit tickets and read 5 quickly aloud (anonymously). Close: "A 49% chance for both days is still a chance, not a promise. In Lesson 8 we will draw a tree diagram to see all four outcomes together."	Cumulative model revision: learners add the multiplication law to the existing model and explain how each piece connects to the phenomenon (records, sample space, addition law, multiplication law). This shows the model growing and prepares for the tree diagram and the decisions of Lesson 9.	Look for: (a) the new panel shows both 0.49 and 0.09 with the correct calculations; (b) learners state that the multiplication law needs independent events; (c) the exit ticket correctly says the two-day event is less likely because both things have to happen (e.g. 'it takes two 70% events and each can fail'); (d) learners link the model to a decision for Wanjiru, noting that 0.49 is not a guarantee. Use the exit tickets to decide which learners need re-teaching before Lesson 8.

D. TEACHER REFLECTION

After the lesson ask yourself: Did learners decide independence before multiplying, or did they multiply automatically? How many still wrote $0.7 + 0.7 = 1.4$ and did my questions help them see why a probability cannot exceed 1? Did the pooled coin and die data convince learners that multiplication fits real results, and what did learners say when their group's relative frequency was not exactly 0.25? Were the predictions from the Predict Phase revisited honestly? Which learners struggled with decimal multiplication (0.7×0.3) rather than with the idea of probability, and what support do they need? Note which questions on the DQB are ready for Lesson 8 (tree diagrams, exactly one rainy day, at least one rainy day) and adjust the grouping so that confident learners support others.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We tossed a coin and rolled a die together and recorded how often we got a head and an even number. We also wrote the four outcomes for two days of weather (RR, RN, NR, NN) with $P(R) = 0.7$ and $P(N) = 0.3$ each day.
What did I learn?	For independent events, $P(A \text{ and } B) = P(A) \times P(B)$. The chance of rain on both days is $0.7 \times 0.7 = 0.49$ and the chance of no rain on both days is $0.3 \times 0.3 = 0.09$. The chance of both events is smaller than the chance of either one alone.
How does this explain the phenomenon?	Wanjiru Chebet can use the multiplication law to turn the KMD one-day forecast of 0.7 into a two-day chance of 0.49. This is not a guarantee, but it tells her that about 49 out of every 100 such pairs of days have rain on both days, which helps her decide whether to plant, wait or irrigate.

LESSON 8 (40 minutes (1 lesson)): Probability Tree Diagrams: Mapping Two Days of Weather

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners draw and use probability tree diagrams to show combined events such as the weather over two days, so that they can multiply along branches, add the outcomes that make an event, and see how forecast numbers are combined for decisions.
Knowledge	Learners will be able to: (1) draw a two-stage tree diagram for two days with R ($P = 0.7$) and N ($P = 0.3$) on each day; (2) write probabilities on branches and multiply along each path using the multiplication law: RR 0.49, RN 0.21, NR 0.21, NN 0.09; (3) add the outcomes that make an event using the addition law: exactly one rainy day = $0.21 + 0.21 = 0.42$, at least one rainy day = $0.49 + 0.21 + 0.21 = 0.91$; (4) check that all final outcomes sum to 1.00 and check 'at least one' with the complement, $1 - 0.09 = 0.91$.
Skills	Developing and using models (NGSS SEP 2): building a tree diagram as a model of two days of weather. Analysing and interpreting data (SEP 4). Using mathematics and computational thinking (SEP 5): multiplying and adding probabilities. Constructing explanations (SEP 6) and engaging in argument from evidence (SEP 7): justifying which branches belong to an event. Communication and collaboration in groups of 5.
Attitudes	Learners appreciate that careful, organised diagrams help people make sensible decisions under uncertainty; show curiosity, accuracy and perseverance when checking that probabilities sum to 1; and value teamwork and respectful challenge when groups compare trees.
Key Inquiry Question	1. How do we determine the probability of an event? 2. How do we apply probability in real life situations?
Purpose in Storyline	Lesson 6 gave us the addition law (and the complement rule) and Lesson 7 gave us the multiplication law for independent events, including $P(RR) = 0.49$ and $P(NN) = 0.09$. In this lesson learners organise both laws in one picture, the tree diagram, for two days of weather in Kericho and the Rift Valley. The complete two-day model (RR 0.49, RN 0.21, NR 0.21, NN 0.09, total 1.00; at least one rainy day 0.91; exactly one 0.42) is the evidence that Lesson 9 will use when the farmer Wanjiru, Captain Otieno and Ms Achieng's county office make decisions.
Safety Notes	Low-risk lesson. Learners use paper, pencils, rulers and, if desired, a coin or bottle top to test the tree. Ask learners to keep coins and bottle tops on desks and not to throw them. Keep walkways clear during the gallery walk. Ensure every learner in a group of 5 has a role.

B. LESSON OVERVIEW

Mr Kamau reminds the class of the puzzle: the KMD forecasts a 70% chance of rain in Kericho and the Rift Valley, yet Wanjiru Chebet, a maize farmer near Eldoret, must plan for more than one day. Learners predict how likely rain is on at least one of two days, then build a tree diagram step by step. They write 0.7 and 0.3 on the branches, multiply along each path (0.49, 0.21, 0.21, 0.09), add outcomes to answer 'exactly one rainy day' (0.42) and 'at least one rainy day' (0.91), and check with the complement $1 - 0.09$. Learners then add what they learned to the Driving Question Board and revise the class model so it can combine probabilities for decisions in Lesson 9.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners sit in groups of 5. Mr	Say: 'Last lesson we met the	Predict-then-Test (Prediction	Look for: (1) learners who give

 VIDEO: Count outcomes using tree diagram Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Probability with general multiplication rule Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Kamau shows the situation: 'Wanjiru Chebet near Eldoret plans her planting over the next two days. The KMD says $P(\text{rain}) = 0.7$ on each day, and the days are independent.' Individually, each learner writes in their notebook a prediction on a 0 to 1 scale (or %): (a) How likely is rain on BOTH days? (b) How likely is NO rain on both days? (c) How likely is rain on AT LEAST one of the two days? They give a one-sentence reason. Groups then compare predictions and agree on one group prediction for (c), written on a sticky note or half-sheet of paper. Learners also try to sketch, however they like, a picture that shows all the things that could happen over the two days.	multiplication law. Today we ask a bigger question: with two days, what are ALL the things that could happen, and how likely is each?' Write on the board: $P(\text{rain}) = 0.7$ each day. Give 2 minutes of silent individual writing. Say: 'Write a number, not just a word, and say why.' Circulate and note sketches. WAIT TIME: after asking 'Is rain on at least one day more likely or less likely than rain on one particular day?', wait 10-15 seconds before taking responses. Cold-call 3 learners from different groups to share their group's prediction for (c) and their reasoning. Do not say which is right. Record the range of predictions on the board (for example 0.7, 0.8, 0.9, 1.0). Ask: 'What kind of picture could help us see every possibility without missing any?'	Gallery): committing to a number before instruction exposes learners' intuitions (for example that two days of 0.7 gives 1.4, or that it is still just 0.7). Seeing the spread of group predictions creates a need to build a model that can settle the question, and the learners' own sketches become the starting point for the tree diagram.	numbers between 0 and 1 rather than values above 1; (2) learners who recall that $P(\text{RR}) = 0.7 \times 0.7$ from Lesson 7 or who add $0.7 + 0.7$ incorrectly; (3) sketches that list outcomes such as RR, RN, NR, NN or that miss RN and NR. Note on a class checklist which groups listed all four outcomes and which listed fewer. Use these to decide which groups need support when drawing branches.
Observe Phase  VIDEO: Count outcomes using tree diagram Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Probability with general multiplication rule Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners build the tree diagram as a guided activity. Step 1: Each group draws Day 1 with two branches from a starting point: R (0.7) and N (0.3), checking that $0.7 + 0.3 = 1$. Step 2: From the end of EACH Day 1 branch, they draw two Day 2 branches, R (0.7) and N (0.3), explaining why the Day 2 numbers are the same whatever happened on Day 1 (days are independent). Step 3: They write the four final outcomes RR, RN, NR, NN at the ends of the paths. Step 4: They multiply along each path using the multiplication law and write the results: $RR = 0.7 \times 0.7 = 0.49$; $RN = 0.7 \times 0.3 = 0.21$; $NR = 0.3 \times 0.7 = 0.21$; $NN = 0.3 \times 0.3 = 0.09$. Step 5: They add the four outcomes and confirm that the total is $0.49 + 0.21 + 0.21 + 0.09 = 1.00$. Step 6 (check against real data): Using a coin (Heads = R,	Model Step 1 on the board slowly, drawing only the first set of branches. Ask: 'Why must the two branches from one point add to 1?' WAIT TIME 10-15 seconds, then cold-call 2 learners. Move to Step 2 and ask: 'Does it matter if Day 1 was rain or no rain when we write the Day 2 branches? Why not?' Cold-call 3 learners; look for the idea of independence from Lesson 5 and Lesson 7. Distribute a prepared blank tree outline to groups that need it. During Step 4 say: 'Multiply ALONG the branches. Say the path out loud, for example: rain, then rain, 0.7 times 0.7.' Walk around and check that each group has the four products. For Step 5 ask: 'Why should the total be exactly 1.00?' Cold-call 2 learners. For Step 6, remind learners that the coin is a fair one with $P(H) = 0.5$, so it is	Scaffolded Model Construction with Structured Data Table: the tree is built branch by branch so that learners see why each number is written where it is. Linking each path to a product and then summing all of them to 1.00 turns two separate laws from Lessons 6 and 7 into one coherent model. Tracing a coin-toss path makes the idea of 'one path through the tree' concrete.	Check that: (1) Day 2 branches start from the end of EACH Day 1 branch (4 final outcomes, not 2 or 3); (2) probabilities are written on branches and products at the ends; (3) each learner can state aloud that the two branches from each point sum to 1; (4) the products are correct (0.49, 0.21, 0.21, 0.09) and sum to 1.00; (5) learners correctly identify that the coin path is only an illustration. Quick check: ask each group 'Which two paths have the same probability and why?' Look for 'RN and NR because both use one 0.7 and one 0.3'.

	Tails = N) or a bottle top, each group does 1 round of 2 tosses (one toss per 'day') to show one path of the tree and records which path it matches. Learners note that a coin is a fair 0.5 and so does not match 0.7, and that it only shows how a path is traced, not the 0.7 and 0.3 values.	used only to follow a path and not to give the 0.7 value. Say: 'Point to the path your two tosses followed.' Have one learner per group point to it on the group tree. Display one correct tree on the board only AFTER groups have finished their own, and ask groups to compare and correct in a different colour.		
Explain Phase  VIDEO: Sample spaces for compound events <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Probability of "at least one" success <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups use their finished tree to answer questions about events. (a) Exactly one rainy day: which paths? RN and NR; add $0.21 + 0.21 = 0.42$ using the addition law (these outcomes are mutually exclusive, because one path cannot be both RN and NR). (b) At least one rainy day: list the paths RR, RN, NR and add $0.49 + 0.21 + 0.21 = 0.91$. (c) Check by complement: the only path with no rain is NN = 0.09, so $1 - 0.09 = 0.91$. (d) Rain on both days = 0.49, and no rain on both days = 0.09. Each learner writes a sentence in context, for example: 'The chance that Wanjiru gets at least one rainy day out of two near Eldoret is 0.91, about 91 in 100 similar two-day periods.' Groups compare their Step 1 predictions with the results and explain, using the tree, why 'at least one rainy day' is higher than 0.7 and why it is not 1.4. Early finishers consider: 'Does 0.49 mean it will rain on both days? What does it mean for Wanjiru?' and write an explanation that reflects uncertainty (it might still stay dry).	Pose each event question one at a time. Say: 'Put your finger on every path that belongs to this event. Then tell me which law you use to combine them.' For (a), ask: 'Which paths give exactly one rainy day? Why do we ADD those two?' WAIT TIME 10-15 seconds, then cold-call 3 learners from different groups. Prompt: 'Multiply along, add across.' For (b), ask: 'How many paths have at least one R?' Cold-call 2 learners; address the misconception that 'at least one' means 'exactly one'. For (c), say: 'Find a quicker way using the complement from Lesson 6.' Cold-call 2 learners and have them show $1 - 0.09 = 0.91$. Ask: 'Why do both methods agree?' Link to total = 1.00. Challenge: 'Our forecast says 0.7 for one day, but at least one rainy day in two is 0.91. Why is it higher?' Take 3 responses and revisit initial predictions. Ask: 'If the day turned out dry, was the forecast wrong?' Take 2 responses and affirm that a 0.09 chance of two dry days can still happen. Keep to the numbers in the established data; do not introduce conditional probability.	Claim-Evidence-Reasoning (CER) with Think-Pair-Share: learners make a claim (for example 'P(at least one rainy day) = 0.91'), cite evidence from the tree (the three paths with R, or the single NN path) and give reasoning with the addition and multiplication laws. Two methods reaching the same answer builds confidence in the model, and comparing with initial predictions shows how the evidence changed their thinking.	Observable indicators: learners circle the correct paths for each event on their trees; they say 'multiply along, add across'; they obtain 0.42 and 0.91 and justify with paths or complement; they write sentences that interpret 0.91 as a long-run chance rather than a guarantee. Use a 3-finger check: 3 = can explain both methods for 'at least one', 2 = can do one method, 1 = unsure. Pair learners with a 1 with a learner who shows 3. Collect exit evidence of misconceptions such as adding $0.7 + 0.7$ or treating 'at least one' as 'exactly one'.
Driving Question Board (DQB) Creation  VIDEO:	Learners return to the class Driving Question Board (DQB) headed 'How is probability applied in real life situations?' Each group reviews the questions from earlier	Say: 'Look at our board. Which questions can the tree diagram help us answer today?' WAIT TIME 10-15 seconds before groups move. Cold-call 3 groups to	Driving Question Board (DQB) Tracking: connecting each piece of evidence to a public question makes progress visible and keeps learners' own questions at the	Check that sticky notes cite specific tree evidence (numbers or paths) rather than general statements. Look for new questions that connect tree

Conditional probability tree diagram example <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Complement Rule for Probability (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	lessons and decides which have been answered by the tree diagram, for example 'How are such numbers combined?' and 'Why is the chance of at least one rainy day higher than 0.7?'. They write a short answer on a sticky note with evidence (the tree, 0.49, 0.21, 0.21, 0.09, 0.42, 0.91) and place it beside the question. They also add at least one NEW question that has emerged, for example: 'What would Captain Otieno's tree look like with a storm included?', 'How does Ms Achieng's office use these numbers?' or 'Should Wanjiru plant if P(RR) is 0.49?'	read a question and the evidence they used. Ask: 'What evidence from the tree supports your answer?' Push for specific numbers. Say: 'Which questions are still open?' Take 2 responses and mark them with a star as the focus for Lesson 9. Make sure the sticky notes are placed on the DQB and sorted into 'answered', 'partly answered' and 'new questions'. Note that decision rules in Lesson 9 are classroom rules and not real KMD rules.	centre. Sorting questions into answered, partly answered and new helps learners see what they know and what they still need in order to link probabilities to decisions.	outcomes to decisions by Wanjiru, Captain Otieno or Ms Achieng. Note learners who ask about storms or thresholds as ready for Lesson 9. Record which groups do not yet link the tree to the phenomenon.
Model Building Phase VIDEO: Probability of a compound event <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Complement Rule for Probability Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners revise the class model of the forecast phenomenon. Individually, each learner updates their own model page with: (1) the one-day model ($S = \{\text{rain, storm, dry}\}$; $P(\text{rain}) = 0.7$, $P(\text{no rain}) = 0.3$ as used in the tree); (2) the two-day tree diagram with branch probabilities and the four outcome probabilities; (3) a short list of rules: 'multiply along the branches, add the outcomes that make the event, all outcomes sum to 1'; (4) the results 'exactly one rainy day = 0.42' and 'at least one rainy day = 0.91'; and (5) a sentence explaining what these numbers tell Wanjiru, for example that over many similar two-day periods about 91 in 100 would have at least one rainy day, though any single pair of days may be unlucky. Groups then do a gallery walk to compare models and leave one 'glow' (what is clear) and one	Say: 'Our model has to explain what happens over two days, not just one. Add the tree to your model and show how it connects to Wanjiru's decision.' Give 8 minutes for individual updating and 5 minutes for the gallery walk. Circulate and ask: 'Which part of your tree shows at least one rainy day?' and 'How does this help Wanjiru, Captain Otieno or Ms Achieng?' WAIT TIME 10-15 seconds before taking answers. Cold-call 3 learners to share one revision they made to their model and why. Remind them that the model is still incomplete because the storm state has not been combined across days and decisions have not yet been made. Preview Lesson 9: 'Next time we use the full model to decide what each person should do and to answer our driving question.'	Cumulative Model Revision with Gallery Walk: updating the same model across lessons shows how new evidence (tree outcomes) extends earlier ideas (relative frequency, sample space, laws). Peer glow/grow feedback encourages learners to justify and refine representations, and writing a sentence in context links the mathematics back to the phenomenon.	Observable indicators: models include a correct tree with branch values 0.7 and 0.3 and outcomes 0.49, 0.21, 0.21, 0.09 summing to 1.00; both event probabilities (0.42 and 0.91) are shown with working; the interpretation sentence treats probability as a long-run chance, not a guarantee. Use a success checklist (tree correct, laws stated, event answers correct, context sentence). Collect model pages from 6 learners to review before Lesson 9 and identify those needing support with 'at least one' or complement thinking.

	'grow' (what to improve) comment. Learners finish by writing one sentence on how a tree diagram helps decision makers combine probabilities.			
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did every group's tree have four final outcomes with Day 2 branches starting from each Day 1 branch? Which learners drew only two or three outcomes, and what support helped? (2) How many learners could explain, in their own words, why we multiply along branches and add across outcomes? (3) Which misconception was most common: adding $0.7 + 0.7$, treating 'at least one' as 'exactly one', or reading 0.49 as a guarantee of rain on both days? (4) Did the complement check ($1 - 0.09 = 0.91$) help learners trust the answer? (5) Were learners able to connect the tree to Wanjiru's decision without conditional probability? (6) What change to wait time, grouping or prepared tree outlines would improve participation in Lesson 9? Note which learners are ready to use the full model, including the 0.1 storm probability, to justify decisions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We drew a two-day tree for rain (R, 0.7) and no rain (N, 0.3) and found the four outcomes RR, RN, NR, NN. Multiplying along the branches gave 0.49, 0.21, 0.21 and 0.09, and these add to 1.00.
What did I learn?	A tree diagram shows all outcomes of combined events. We multiply along branches (multiplication law) and add the outcomes that make an event (addition law). Exactly one rainy day = $0.21 + 0.21 = 0.42$, and at least one rainy day = $0.49 + 0.21 + 0.21 = 0.91$, which equals $1 - 0.09$.
How does this explain the phenomenon?	Even though the forecast for one day is 0.7, the chance of at least one rainy day in two is higher, 0.91, because there are more ways to get rain. The tree lets Wanjiru, Captain Otieno and the county office combine probabilities, but the results are long-run chances, so a single pair of days can still be dry, with probability 0.09.

LESSON 9 (80 minutes (2 x 40-minute periods)): Applying probability to real decisions: advising the farmer, the pilot and the county office

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use their complete probability model (relative frequency, sample space, addition law, multiplication law, complement and tree diagrams) to justify decisions for Wanjiru Chebet, Captain Otieno and Ms Achieng's county disaster office, finalise the Driving Question Board and write a Final Explanation answering: How is probability applied in real life situations?
Knowledge	Learners will explain that a forecast of 70% rain is a long-run measure of likelihood. It comes from past records (Kericho 14/20 = 0.7) and models. It does not guarantee the outcome on one day. They will use $P(\text{rain}) = 0.7$, $P(\text{storm}) = 0.1$, the day-state space $S = \{\text{rain, storm, dry}\}$ and the two-day tree (RR 0.49, RN 0.21, NR 0.21, NN 0.09; at least one rainy day 0.91; exactly one 0.42) to compare probabilities with class decision thresholds. These thresholds are classroom rules, not real KMD rules.
Skills	Calculating and comparing probabilities with thresholds; selecting the right law (addition, multiplication or complement) for a decision; reading and using a tree diagram; Developing and Using Models (NGSS SEP 2): comparing the Lesson 1 model with the final model; Engaging in Argument from Evidence (SEP 7): justifying a recommendation with numbers; Obtaining, Evaluating and Communicating Information (SEP 8); using mathematics and computational thinking (SEP 5); collaboration and communication.
Attitudes	Appreciate that probability helps people in farming, aviation and disaster management make sensible decisions under uncertainty; show respect for others' reasoning in group discussion; be willing to revise ideas when evidence changes; accept that a good decision can still have an unlucky result on one day.
Key Inquiry Question	1. How do we determine the probability of an event? 2. How do we apply probability in real life situations?
Purpose in Storyline	This is the final lesson. In Lessons 1-2 learners estimated likelihood from relative frequency (coin tosses, pooled 42/80, Kericho 14/20). In Lesson 3 they listed sample spaces and events. In Lesson 4 they found theoretical probability and the day-state probabilities 0.6, 0.1 and 0.3. In Lesson 5 they classified events as mutually exclusive or independent. In Lesson 6 they used the addition law and complement. In Lesson 7 they used the multiplication law. In Lesson 8 they drew the two-day tree. Here they bring all of this together to answer the original puzzle: why do farmers, pilots and cities trust and act on numbers that do not guarantee the outcome? This lesson teaches no new content.
Safety Notes	Low risk, paper-based lesson. Use coins or bottle-tops only on desks, not thrown across the room. Remind learners to be kind when peers share incorrect ideas. Make clear that the thresholds are classroom decision rules and not real KMD, aviation or county rules; real decisions use official guidance.

B. LESSON OVERVIEW

Learners return to the March morning forecast of 70% chance of rain in Kericho and the Rift Valley. First they individually predict what each decision-maker should do and whether a 70% forecast can be trusted. Then, in groups of 5, they observe evidence by working through three decision cards using the full model: Wanjiru near Eldoret (two-day tree, thresholds 0.49 and 0.9), Captain Otieno ($P(\text{storm}) = 0.1$ against the 0.1 threshold) and Ms Achieng's office ($P(\text{rain}) = 0.7$ or $P(\text{storm}) = 0.1$). In the Explain phase they write and critique a Final Explanation using claim, evidence and reasoning. The class then finalises the DQB, sorting questions into answered and still open. Finally learners compare their Lesson 1 model with their final model to show how their thinking changed.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Addition rule for probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Probabilities of compound events <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at the original forecast card: 'KMD: 70% chance of rain in Kericho and the Rift Valley; 10% chance of storm.' Individually, in 5 minutes, they write predictions in their books: (a) What should Wanjiru do near Eldoret: plant today, wait, or irrigate? (b) Should Captain Otieno worry about the flight to Kisumu? (c) Should Ms Achieng's office prepare drainage? (d) If it stays dry today, was the 70% forecast wrong? Why or why not? They then turn to a partner and compare answers for 2 minutes, marking any prediction they are unsure about with a '?'.	Display the March morning forecast and the three characters (Wanjiru Chebet, Captain Otieno, Ms Achieng). Say: 'On our first day you saw this forecast and were puzzled. Before we use everything we have learned, write what you think each person should do and why.' Give 5 minutes of silent writing, then circulate and note 3 or 4 interesting predictions. After pair talk, cold-call 4 learners with: 'What did your partner predict that was different from yours?' WAIT TIME 10-15 seconds before each cold-call. Do not correct answers yet. Write 3 predictions on the board under the headings Farmer, Pilot and County. Say: 'We will test these predictions with the evidence we have gathered over eight lessons.'	Predict-Compare-Revise: committing to a written prediction before the evidence work makes learners aware of their current thinking. This creates a clear 'before' point to revise later, and it brings the misconception 'a forecast is wrong if it does not happen' into the open.	Look for: (1) whether predictions mention numbers such as 0.7 or 0.1 rather than just 'it will rain'; (2) whether learners say a forecast is 'wrong' if the day is dry (a misconception to track); (3) whether any learner already links the forecast to past records. Record the count of learners holding the misconception, to compare with the end of the lesson.
Observe Phase  VIDEO: Count outcomes using tree diagram Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Probability with general multiplication rule <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES	In groups of 5, learners work through three decision cards, with roles of reader, calculator, tree-drawer, checker and reporter, rotating every card. Card 1, Wanjiru near Eldoret: they redraw the two-day tree (R 0.7, N 0.3) and read off $P(RR) = 0.49$, $P(\text{at least one rainy day}) = 1 - 0.09 = 0.91$ and $P(\text{exactly one}) = 0.42$. They compare with the class rule: plant if $P(\text{rain on each of the next two days}) \geq 0.49$ is considered good, or if $P(\text{at least one rainy day in two}) \geq 0.9$. Card 2, Captain Otieno: they use $S = \{\text{rain only, storm, dry}\}$ with 0.6, 0.1 and 0.3 (sum = 1.0) and compare $P(\text{storm}) = 0.1$ with the rule 'delay or reroute if $P(\text{storm}) \geq 0.1$ '. Card 3, Ms Achieng: they use the addition law	Hand out the cards and state: 'These thresholds are classroom decision rules for our practice, not real KMD rules.' Circulate and ask each group: 'Which law did you use, and how do you know the events are independent or mutually exclusive?' (backs Lessons 5-7). If groups struggle with the tree, say: 'Remind me, what do we do along a branch? What do we do across outcomes?' Prompt on Card 1: 'Why is 0.91 a better reason to plant than 0.49?' Prompt on Card 2: 'The probability is exactly equal to the threshold. What does that mean for the pilot?' Prompt on Card 3: 'Which of the two conditions is met? Both?' Give groups 20 minutes. Then call 3 groups to report, one per card.	Jigsaw-style evidence table with Using Mathematics (SEP 5): the same model (laws and tree) is applied to three different contexts, so learners see that one set of ideas supports many decisions. The table makes the link between calculation and decision explicit, which is the core of applying probability.	Check that groups: (1) reach $P(RR) = 0.49$, $P(\text{at least one}) = 0.91$, $P(\text{exactly one}) = 0.42$; (2) add $0.6 + 0.1 = 0.7$ and recognise that rain for the farmer includes storm; (3) correctly say 0.1 meets the pilot threshold; (4) state the decision as a comparison of a probability with a threshold. Common errors to note: adding instead of multiplying along a branch; using $0.7 + 0.7$ for two days; forgetting that tree outcomes sum to 1. Use a quick tally on the board of how many groups got each answer right on the first attempt.

for similar readings	$P(\text{rain only or storm}) = 0.6 + 0.1 = 0.7$, and compare 0.7 and 0.1 with 'prepare drainage and warn residents if $P(\text{rain}) \geq 0.7$ or $P(\text{storm}) \geq 0.1$ '. They record their evidence in a table: Person Probability used Law or tool Threshold Decision.	WAIT TIME 10-15 seconds after each question. Cold-call 2 more learners to restate the group's decision in their own words.		
Explain Phase  VIDEO: Addition rule for probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Sample spaces for compound events <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Individually, learners write a Final Explanation of 8-10 lines using Claim-Evidence-Reasoning that answers: 'Why do Wanjiru, Captain Otieno and the county office act on a 70% forecast even though it does not guarantee rain?' Sentence starters: 'My claim is...', 'The evidence is...' (Kericho 14/20 = 0.7; tree 0.49 and 0.91; $P(\text{storm}) = 0.1$; $0.6 + 0.1 = 0.7$), 'This means...'. They must include: (1) where 0.7 comes from (past records and models, relative frequency); (2) that it means about 7 in every 10 similar days in the long run; (3) how laws and the tree combine probabilities for two days; (4) why a decision can be good and still be unlucky on one day. They swap with a partner and use a checklist to give one star and one suggestion. They then discuss in the whole class: 'If Wanjiru plants and the next two days are dry, was her decision bad?'	Say: 'You are now the experts. Your job is to explain to a neighbour who thinks forecasts are useless.' Share the CER checklist on the board. Give 10 minutes of silent writing, then 5 minutes for peer review. Circulate and push with: 'Show me the number that supports that sentence.' and 'Which lesson gave you that idea?' Use cold-calls: choose 3 learners to read one sentence of evidence. Then pose the unlucky-day question and use WAIT TIME of 15 seconds. Cold-call 3 learners. Draw out the idea: 'A decision based on probability is judged over many days, not on one day.' Link to cost of being wrong: planting early risks losing seed if dry, while waiting risks missing the rains. Pilot: storm risk affects safety. County: flood risk to residents.	Claim-Evidence-Reasoning with peer critique (Engaging in Argument from Evidence, SEP 7): writing forces learners to connect numbers to reasoning, and peer review exposes gaps such as stating a decision with no probability supporting it.	Use a 3-level rubric: Level 3 uses relative frequency (14/20), at least two laws or the tree with correct numbers, and explains long-run meaning and the unlucky-day idea; Level 2 has correct numbers but weak reasoning; Level 1 says only 'it will probably rain'. Collect 5 explanations to review. Look for correct use of the words 'chance' (informal) and 'probability' (formal). Ask: 'Who changed their answer on whether a dry day makes the forecast wrong?' and compare with the Predict phase tally.
Driving Question Board (DQB) Creation  VIDEO: Count outcomes using tree diagram Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners bring the sticky notes of questions they posted over Lessons 1-8 (for example 'Why 70% and not 100%', 'How do they get the number?', 'Can we add the days?', 'What if there is a storm too?'). In their groups, they sort questions into three zones: Answered (with evidence and the lesson where we answered it), Partly answered, and New questions. They write a short	Open the DQB and say: 'Look at the questions we asked on Day 1. Which can you answer now? Show me the evidence.' Give groups 8 minutes for sorting. Ask: 'Which lesson helped you answer this one?' Then facilitate a 4-minute whole-class synthesis. Cold-call 4 learners, one per zone or question type. WAIT TIME 10-15 seconds. Model a strong answer: 'Probability is applied by measuring how likely	Driving Question Board review (public record of learning): sorting questions by status shows learners how far their understanding has grown and keeps unanswered questions visible, which builds ownership and a sense of progress.	Observe whether groups match each question to correct evidence (for example, 'Why 70%?' matched to Kericho 14/20 = 0.7). Count how many questions each group moves to Answered. Note whether new questions are meaningful and linked to the phenomenon. Listen for correct vocabulary: relative frequency, sample space, independent, mutually exclusive, complement.

 EXERCISE: Two-way tables, Venn diagrams, and probability <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	answer on a new note for each Answered question and stick it next to the original. Finally, as a class they write the answer to the driving question 'How is probability applied in real life situations?' in one or two sentences on the DQB, and each group adds one new question to carry forward (for example, 'How do meteorologists build their models?' or 'What if the days are not independent?').	events are from past records and rules, then comparing those probabilities with the cost of being wrong to decide.' Accept genuine open questions and explain that they are a sign of good science. Link back to KICD key inquiries: 'How do we determine the probability of an event?' (relative frequency and theoretical probability) and 'How do we apply probability in real life situations?' (decisions for the farmer, pilot and county).		
Model Building Phase  VIDEO: Probability of a compound event <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Complement Rule for Probability (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners take out the model they drew in Lesson 1 (their first idea of what '70% chance of rain' means and how a forecast is made). On a fresh page, they draw a final model in a flowchart or concept map: past records (Kericho 14/20) -> relative frequency 0.7 -> forecast $P(\text{rain}) = 0.7$ -> sample space $S = \{\text{rain only, storm, dry}\}$ (0.6, 0.1, 0.3) -> laws (addition, multiplication, complement) and the tree diagram -> compare with threshold and cost of being wrong -> decision (Wanjiru, Captain Otieno, Ms Achieng) -> long-run outcome. They label each arrow with the law or idea used. Then they write a 'Then and Now' box with 3 sentences: 'In Lesson 1 I thought... Now I think... because...'. Some learners volunteer to present.	Say: 'Place your first model next to your new one. What is the biggest change?' Give 12 minutes. Circulate and ask: 'Where does the number 0.7 come from in your model? Where is the storm 0.1? Where does the tree fit?' Remind that the model should explain why the forecast is useful even though it is not certain. Select 3 learners to present, choosing different features (data source, laws, decision). After each, cold-call 2 listeners to say one thing they would add. WAIT TIME 10-15 seconds. Close with: 'Probability does not tell us what will happen tomorrow; it tells us how to choose well when we cannot know. Wanjiru, Captain Otieno and Ms Achieng trust the numbers because they come from many past days and obey rules we can check.' Collect models and Final Explanations as a record of learning.	Cumulative model revision with Then-and-Now reflection (Developing and Using Models, SEP 2): comparing the Lesson 1 model with the final model makes the growth in understanding visible and helps learners see that models are revised as evidence grows.	Check that the final model includes: (1) data source linked to relative frequency; (2) sample space with probabilities summing to 1; (3) at least two laws or the tree; (4) a decision step comparing probability with a threshold or cost; (5) the idea that results vary on one day. Check the Then-and-Now sentences for specific changes (for example, from 'forecast is wrong if dry' to 'forecast is a long-run chance'). Record how many learners moved from the misconception to the accepted idea.

D. TEACHER REFLECTION

After the lesson, consider: (1) How many learners moved away from the idea that a forecast is wrong when a dry day happens? What evidence from their Final Explanations shows this? (2) Which law or tool (addition, multiplication, complement, tree) did learners find hardest to select independently, and which earlier lesson

might need revisiting? (3) Did learners treat the classroom thresholds correctly as practice rules and not real KMD rules? (4) Did every learner contribute during group roles, or did some stay passive? How could you adjust the roles next time? (5) Which DQB questions stayed open, and how could they feed the next unit? (6) Did the Kenyan contexts (Wanjiru near Eldoret, Captain Otieno at JKIA, Ms Achieng in Nairobi, Kericho records) help learners see how probability is applied in real life?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the forecast of 70% rain in Kericho matches the past record of 14 rainy days out of 20 ($14/20 = 0.7$), that the day states have probabilities rain only 0.6, storm 0.1 and dry 0.3 (sum 1.0), and that the two-day tree gives RR 0.49, RN 0.21, NR 0.21, NN 0.09 (total 1.00), with at least one rainy day 0.91 and exactly one 0.42. Comparing these with the class thresholds led to different decisions for Wanjiru, Captain Otieno and Ms Achieng.
What did I learn?	We learned that probability is a number from 0 to 1 that measures how likely an event is. It can be estimated from relative frequency or found by theoretical probability. We learned to list sample spaces, to use the addition law for mutually exclusive events, the multiplication law for independent events, the complement rule and tree diagrams. We also learned to compare probabilities with a decision threshold and with the cost of being wrong.
How does this explain the phenomenon?	A forecast of 70% chance of rain does not guarantee rain. It means that in a long run of similar days about 7 out of every 10 have rain, worked out from past records and models. Because these numbers come from many past trials, Wanjiru can prepare to plant and keep irrigation as backup, Captain Otieno can plan for a possible 10% storm and check again before take-off, and the county office can prepare drainage and warn residents. A decision based on probability may be unlucky on one day, but it is the best choice over many days. This is how probability is applied in real life situations.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.